Orthogonal Subspace Projection Using Data Sphering and Low-Rank and Sparse Matrix Decomposition for Hyperspectral Target Detection

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***Abstract*— Orthogonal subspace projection (OSP) has been widely used in many applications for hyperspectral data exploitation. However, its performance is sensitive to its used prior target knowledge, which is significantly affected by target background (BKG). To resolve this issue, this article develops three approaches to extend OSP in improving its performance. One is data sphering which can suppress BKG via removing the first- and second-order data statistics. Another takes advantage of a recently developed low-rank and sparse matrix decomposition (LRaSMD) to separate BKG and target signal sources in two subspaces characterized by the low-rank matrix and sparse matrix, respectively, and then annihilates BKG via the low-rank matrix, referred to as BKG-annihilated OSP (BA-OSP). A third approach combines data sphering and LRaSMD to further improve OSP over target detectability and BKG suppressibility. Experiments show that implementing OSP in conjunction with data sphering and LRaSMD significantly improves OSP in target detection and BKG suppression, and also performs as well as a widely used constrained energy minimization (CEM)-based subpixel target detection.**

***Index Terms*— Constrained energy minimization (CEM), ensemble-cascaded CEM, go decomposition (GoDec), low-rank and sparse matrix decomposition (LRaSMD), orthogonal subspace projection (OSP).**

I. Introduction

ORTHOGONAL subspace projection (OSP) has found its great potential in many applications in hyperspectral data exploitation [1]–[4]. A brief OSP history can help to understand how OSP has advanced its current status and its evolution in various stages from its earliest application in dimensionality reduction up to the work presented in this article and may provide future and potential research directions.

OSP was first envisioned and developed for hyperspectral classification and dimensionality reduction [5] with its extended and detailed discussions in [6] along with its recursive version, called recursive OSP (ROSP) [7]. It was also extended to the target-constrained interference-minimized filter (TCIMF) [8] and signal-decomposed and interference-annihilated (SDIA) model [9]. Its close relationship with least-squares error-based spectral unmixing was also studied in [10] and [11]. In addition, its applications were also explored for multispectral imaging in [12] and medical imaging in [13] and [14]. By taking advantage of OSP automatic target generation process (ATGP) was later developed as an OSP-based unsupervised target detection in [15] with its application to endmember finding where ATGP was used as an unsupervised OSP to find endmembers [16] along with its recursive version developed in [17]. Recently, OSP and ATGP were further extended to progressive band processing [18], [19]. Most recently, the concept of OSP was found to be applicable to virtual dimensionality (VD) estimation [20].

Technically speaking, OSP is a mixed multiple-target detection technique, which detects the abundance fractions of a set of p multiple targets, $\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_p$ of interest for target detection. Unlike signal detection in communications [21] where only one single target signal of interest to be detected, hyperspectral target detection deals with a signal which is indeed a mixed-signal, generally specified by $\mathbf{M}\alpha$ with $\mathbf{M} = [\mathbf{m}_1\mathbf{m}_2, \ldots, \mathbf{m}_p]$ being the signature matrix and α being a p-dimensional vector specified by $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_p)^T$. So, if we directly perform the signal detection theory, the detected signal will be the mixed-signal $\mathbf{M}\alpha$, not any particular signal $\mathbf{m}_j$. OSP resolves this issue by separating a signal of interest to be detected, say $\mathbf{m}_p$ as the desired target signature, and the remaining (p-1) signature, $\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_{p-1}$ as the undesired target signatures. With this formulation, OSP performs a two-stage detection on a data sample $\mathbf{r}$ by first annihilating the undesired target signatures $\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_{p-1}$ via an orthogonal subspace projector given by $P_{\mathbf{U}}^{\perp}$ given by

$$P_{\mathbf{U}}^{\perp} = \mathbf{I} - \mathbf{U}\mathbf{U}^{\#} = \mathbf{I} - \mathbf{U}(\mathbf{U}^T\mathbf{U})^{-1}\mathbf{U}^T \tag{1}$$

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where $\mathbf{U} = [\mathbf{m}_1\mathbf{m}_2, \ldots, \mathbf{m}_{p-1}]$ is an undesired target signature matrix formed by $\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_{p-1}$and $\mathbf{U}^{\#}$ is the pseudo-inverse of $\mathbf{U}$ given by $(\mathbf{U}^T\mathbf{U})^{-1}\mathbf{U}^T$. It is then followed by a traditional signal detector from the signal detection theory in the second stage to perform a matched filter

$$\delta^{\mathrm{OSP}}(\mathbf{r}) = \mathbf{d}^T\left(P_{\mathbf{U}}^{\perp}\mathbf{r}\right) \tag{2}$$

with $\mathbf{d} = \mathbf{m}_p$ used as the desired signal to match the $\mathbf{U}$-annihilated data sample $P_{\mathbf{U}}^{\perp}\mathbf{r}$.

While its success has been evidenced by many reports in the literature, OSP also has a limitation on its utilization due to its sensitivity to prior target knowledge, desired target signature $\mathbf{d}$ and the undesired target signature matrix $\mathbf{U}$. Since a hyperspectral image generally has a very complex background (BKG), finding an appropriate $\mathbf{U}$ to include BKG signatures as undesired target signatures is very challenging. Interestingly, the issue in BKG suppression has not received much attention in many applications over the past years. Until recently, this BKG issue has been brought to considerable interests in anomaly detection such as [3, Chapter 5] and [22]–[25] but it still remains largely unnoticed in target detection [26] and so does in classification [27]–[28].

To mitigate this problem, another approach, called constrained energy minimization (CEM), developed in [29] inverted the sample correlation matrix $\mathbf{R}$ with no need of knowing $\mathbf{U}$ to do what OSP does but effectively suppressed BKG similar to OSP using $P_{\mathbf{U}}^{\perp}$ to annihilate $\mathbf{U}$. As a consequence, CEM significantly improved OSP in target detection as shown in [1] and [31]. It may, therefore, mislead to a misconception that OSP may not be able to compete against CEM in target detection. This article shows otherwise as long as BKG is appropriately being taken care of via $\mathbf{U}$.

Upon implementing OSP specified by (2), an issue arises in using $\mathbf{U}$ which must be known *a priori*. However, it is discovered that finding such a set of undesired target signatures directly from the data did not work effectively because BKG is unknown and $\mathbf{U}$ could not faithfully represent BKG. So, an approach similar to the idea of CEM using $\mathbf{R}^{-1}$ to suppress BKG must be sought.

Surprisingly, performing $\mathbf{R}^{-1}$ is similar to data sphering in signal processing [2]. Generally speaking, BKG and target signals can be modeled and also characterized by second-order statistics and high-order statistics of the data to be processed, respectively. With these assumptions, two approaches can be used to suppress BKG without knowing BKG knowledge. One is data sphering which removes the first- and second-order data statistics to suppress BKG. In this case, the resulting sphered data will still retain target signal sources, which are assumed to be characterized by higher order statistics. Accordingly, the target detectability of OSP can be enhanced by operating OSP on such sphered data. Another approach is derived from a recently developed low-rank and sparse matrix decomposition (LRaSMD) model in [32]–[43]. Its idea is to specify BKG and target signal sources by the low-rank matrix $\mathbf{L}$ and sparse matrix $\mathbf{S}$ separately. In doing so, it uses go decomposition (GoDec) developed in [34] to decompose a data matrix $\mathbf{X}$ into a linear mixture of a low-rank matrix component $\mathbf{L}$ and a sparse matrix component $\mathbf{S}$ as $\mathbf{X} = \mathbf{L}+\mathbf{S}+\mathbf{N}$ with $\mathbf{N}$ assumed to be noise and considered as a small data perturbation. By virtue of LRaSMD, BKG can be characterized by the low-rank matrix $\mathbf{L}$, while target signal sources (i.e., $\mathbf{d}$ and $\mathbf{U}$ used in OSP) can be specified by the sparse matrix, $\mathbf{S}$. In this case, the $\mathbf{U}$ used in $P_{\mathbf{U}}^{\perp}$ in (1) can be then replaced by $\mathbf{L}$ to annihilate BKG via $P_{\mathbf{L}}^{\perp}$. Furthermore, the p target signals of interest, $\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_p$ used by OSP are assumed to be in the target subspace specified by the sparse matrix $\mathbf{S}$. Therefore, OSP can be extended by including a BKG annihilator, $P_{\mathbf{L}}^{\perp}$ as a pre-processing step prior to the implementation of OSP.

There are three ways to extend OSP as follows.

1) It first applies data sphering to a data matrix $\mathbf{X}$ and then performs OSP on the sphered data matrix, $\hat{\mathbf{X}}$.
2) It first applies GoDec to $\mathbf{X}$ to produce LRaSMD: $\mathbf{X} = \mathbf{L} + \mathbf{S} + \mathbf{N}$ with BKG specified by $\mathbf{L}$ and target signal subspace specified by either $\mathbf{S}$ or $\mathbf{L} + \mathbf{S}$, and then performs OSP with $\mathbf{d}$ and $\mathbf{U}$ in the subspace generated by $\mathbf{S}$ or $\mathbf{L} + \mathbf{S}$, while using $P_{\mathbf{L}}^{\perp}$ to annihilate BKG.
3) It applies LRaSMD in conjunction with data sphering to implement OSP.

Once the above OSP-based target detection is developed, a following issue is how to evaluate the performance of a detector. A commonly used evaluation tool for target detection including is to calculate the area under a receiver operating characteristic (ROC) curve based on detection probability, $\mathrm{P_D}$ versus false alarm probability, $\mathrm{P_F}$, referred to as AUC, as a measure to assess target detectability. Unfortunately, it has been shown in [44]–[47] that using the 2-D ROC curve of $\mathrm{P_D}$ versus $\mathrm{P_F}$ can only evaluate the effectiveness of a target detector [22]–[26], but not its target detectability and BKG suppressibility. Similarly, in hyperspectral image classification (HSIC) the overall accuracy and average accuracy which have been widely used for classification measures have the same issue since BKG is exclusively removed from classification in HSIC. This was shown in [27] and [28] where precision (also known as user's accuracy) was introduced as a criterion to measure the effect of BKG on classification. To resolve this issue a 3-D ROC-based analysis was first developed in [44] and later extended to various applications in [45] with more details provided in [2,Chapter 3]. Most recently, such a 3-D ROC analysis has been applied to HSIC [46] as well as target detection in [47]. Specifically, [47] derived several 3-D ROC curve-based detection measures that can jointly measure target detectability and BKG suppressibility of a target detector in more effective means, a task that cannot be accomplished by the traditional 2-D ROC analysis. This article takes advantage of the 3-D ROC curve-derived detection measures in [47] to conduct a comparative performance analysis on target detection.

Several novelties can be summarized below.

1) The first and foremost novelty is the introduction of LRaSMD into OSP to eliminate BKG via $P_{\mathbf{L}}^{\perp}$ as a pre-processing step to improve its target detectability. The novel idea is to use the $\mathbf{L}$ obtained from the GoDec-generated LRaSMD to derive BKG-annihilated OSP (BA-OSP) to implement a BKG annihilator $P_{\mathbf{L}}^{\perp}$ derived from (1).

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2) Data sphering can be also used as a pre-processing step to suppress BKG as $P_{\mathbf{L}}^{\perp}$ does by removing the first- and second-order data statistics so that when OSP is operated on the sphered data, a large part of BKG may be then removed prior to OSP.
3) A third novel idea is to combine data sphering as a preprocessing step with LRaSMD to further reduce BKG effects.
4) A fourth novel idea is to make use of different combinations of the low-rank matrix $\mathbf{L}$ and sparse matrix $\mathbf{S}$ to specify $\mathbf{d}$ and $\mathbf{U}$ in (1) to derive various versions of BA-OSP for target detection.
5) Despite the fact that the 3-D ROC detection measures used in this article have their origins in [47], a fifth novel idea is the use of BA-OSP to demonstrate the effectiveness of these detection measures in addressing the BKG suppression issue via extensive experiments, while the traditional 2-D ROC analysis fails in doing so.

The remainder of this article is organized as follows. Section II provides a brief review of GoDec and its related works. Section III derives a data sphered OSP (DSOSP). Section IV develops various versions of BA-OSP using different combinations of GoDec-generated the low-rank matrix, $\mathbf{L}$ and sparse matrix, $\mathbf{S}$. Section V combines data sphering and GoDec-generated LRaSMD to jointly implement OSP. Section VI provides a brief discussion of CEM that will be used for a comparative study and analysis. Section VII develops several 3-D ROC curve-derived detection measures. Section VIII describes the image data to be used for experiments. Section IX performs extensive experiments and discusses their results. Section X conducts a complexity analysis among other most recent developed target detection techniques. Finally, Section XI summarizes contributions and future studies.

II. GoDec and Its Related Works

The history of OSP was discussed in the introduction. This section reviews the key development of the GoDec algorithm and its related works.

Since Zhou and Tao [34] developed the GoDec to solve the low-rank and the sparse matrix approximation, GoDec has evolved into different variants to address the issue of separating BKG from interesting objects. In particular, robust principal component analysis (RPCA) by Candès *et al.* [48] develops a data model $\mathbf{X} = \mathbf{L} + \mathbf{S}$, where $\mathbf{L}$ and $\mathbf{S}$ are a low-rank matrix and a sparse matrix, respectively. It was first proposed by Candès *et al.* [48] where a given data $\mathbf{X}$ is assumed to be of low inherent dimensionality. In contrast to classific principal component analysis (PCA), RPCA can cope with the noise of extremely high amplitude using principal component pursuit (PCP) estimate that solves

$$\min \|\mathbf{L}\|_* + \lambda \|\mathbf{S}\|_{l1} \text{ s.t. } \mathbf{X} = \mathbf{L} + \mathbf{S} \tag{3}$$

where $\|\mathbf{L}\|_*$ is defined by the nuclear norm.

Inspired by compressive sensing (CS) [49], randomized matrix approximation [50], and RPCA, GoDec develops its theory to manipulate the low-rank approximation and sparse presentation. It first projects an input dense matrix on its columns space, $\mathbf{Y}_1 = \mathbf{X}\mathbf{A}_1$ and then its rows space, $\mathbf{Y}_2 = \mathbf{X}^T\mathbf{A}_2$, respectively, where $\mathbf{A}_1$, $\mathbf{A}_2$are random matrices. The projection process is a so-called bilateral random projection (BRP). This process performs like Halko's randomized approximate matrix decomposition [50] so that the low-rank matrix $\mathbf{L}$ is given by

$$\mathbf{L} = \mathbf{Y}_1(\mathbf{A}_2\mathbf{Y}_1)^{-1}\mathbf{Y}_2 \tag{4}$$

After obtaining the low-rank approximation, GoDec decomposes $\mathbf{X}$ into a triplet-component modeled by

$$\mathbf{X} = \mathbf{L} + \mathbf{S} + \mathbf{N} \text{ s.t rank}(\mathbf{L}) \le r, \quad \text{card}(\mathbf{S}) \le k. \tag{5}$$

Recall that RPCA uses the $\mathbf{X} = \mathbf{L} + \mathbf{S}$ in a similar manner. But the difference between RPCA and GoDec is that $\mathbf{S}$ in (5) contains only sparse signals, whereas $\mathbf{S}$ in RPCA is a matrix containing noise.

As noted above, both the randomized power iteration algorithm [50] and GoDec [34] provide a means of overcoming the singular values with slow decay operating on $\mathbf{X}^{\sim} = (\mathbf{X}\mathbf{X}^T)^q\mathbf{X}$ rather than the original input matrix $\mathbf{X}$. In a randomized power iteration algorithm, it uses $\mathbf{X}^{\sim}$ to perform random sampling and then to obtain the sampling matrix. On the other hand, GoDec uses BRP to solve the following low-rank approximation

$$\mathbf{L} = \mathbf{Q}_1\left[\mathbf{R}_1\left(\mathbf{A}_2^T\mathbf{Y}_1\right)^{-1}\mathbf{R}_2^T\right]^{1/(2q+1)}\mathbf{Q}_2^T \tag{6}$$

where $\mathbf{Q}_1, \mathbf{Q}_2, \mathbf{R}_1, \mathbf{R}_2$ are QR decomposition of $\mathbf{Y}_1, \mathbf{Y}_2$. In addition, the model in (5) can be further solved by minimizing the following decomposition error:

$$\begin{aligned} &\min_{\mathbf{L}_{N\times L}, \mathbf{S}_{N\times L}} ||\mathbf{X}_{N\times L} - \mathbf{L}_{N\times L} - \mathbf{S}_{N\times L}||_F^2 \\ &\text{s.t. rank}(\mathbf{L}_{N\times L}) \le m \text{ and cardinality}(\mathbf{S}_{N\times L}) \le jN \end{aligned} \tag{7}$$

where $||\cdot||_F$ is the Frobenius matrix norm. Since (7) does not have an analytical solution, it must rely on a numerical algorithm, which is GoDec to find solutions by the following iterative processes:

$$\mathbf{L}_{N\times L}^{(t)} = \arg\min_{\text{rank}(\mathbf{L}_{N\times L})\le m} ||\mathbf{X}_{N\times L} - \mathbf{L}_{N\times L} - \mathbf{S}_{N\times L}^{(t-1)}||_F^2 \tag{8}$$

$$\mathbf{S}_{N\times L}^{(t)} = \arg\min_{\text{cardnality}(\mathbf{S}_{N\times L})\le jN} ||\mathbf{X}_{N\times L} - \mathbf{L}_{N\times L}^{(t-1)} - \mathbf{S}_{N\times L}||_F^2. \tag{9}$$

The details of the above process to find the low-rank and the sparsity matrix can be summarized as follows.

As described in the above implementation of GoDec, four parameters, m, k, ε, q must be determined in advance. Unfortunately, finding appropriate these values for these parameters is very challenging and GoDec does not provide any guideline for determining these values. So, in applications, these values are generally determined empirically. In a recent work [51], GoDec was further extended to OSP-based GoDec (OSP-GoDec) which resolved this issue by using a concept of VD in [4], [52]–[54] and the minimax singular value decomposition (MXSVD) developed in [55]–[57].

It is worth noting that there are several significant differences between the work presented in this article and the work in [51]. The first significant difference is their different

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Algorithm 1 GoDec

Input: $\mathbf{X}_{N\times L}$ input data matrix
m: the rank of the low-rank matrix $\mathbf{L}$
k: sparsity cardinality for sparsity matrix $\mathbf{S}$
ε : pre-determined the error
q: power scheme
Output:
$\mathbf{L}_{N\times L}$: Low-rank matrix
$\mathbf{S}_{N\times L}$: Sparsity matrix

Initialize: $\mathbf{L}^{(0)}_{N\times L}=\mathbf{X}_{N\times L}$, $\mathbf{S}^{(0)}_{N\times L}=\mathrm{O}_{N\times L}$, $t=0$
While $||\mathbf{X}_{N\times L}-\mathbf{L}_{N\times L}-\mathbf{S}^{(t-1)}_{N\times L}||^2_F/\ \|\mathbf{X}_{N\times L}\|^2_F>\varepsilon$ **Do**
$t=t+1$;
$\mathbf{L}^{\sim}=[(\mathbf{X}_{N\times L}-\mathbf{S}^{t-1}_{N\times L})(\mathbf{X}_{N\times L}-\mathbf{S}^{t-1}_{N\times L})^T]^{\mathbf{q}}(\mathbf{X}_{N\times L}-\mathbf{S}^{t-1}_{N\times L})$
$\mathbf{Y}_1=\mathbf{L}^{\sim}\mathbf{A}_1, \mathbf{A}_2=\mathbf{Y}_1$;
$\mathbf{Y}_2=(\mathbf{L}^{\sim})^T\mathbf{Y}_1=\mathbf{Q}_2\mathbf{R}_2, \mathbf{Y}_1=\mathbf{L}^{\sim}\mathbf{Y}_2=\mathbf{Q}_1\mathbf{R}_1$;
$\mathbf{L}^{(t)}_{N\times L}=\mathbf{Q}_1[\mathbf{R}_1(\mathbf{A}_2^T\mathbf{Y}_1)^{-1}(\mathbf{R}_2)^T]^{1/(2q+1)}\mathbf{Q}_2^T$;
$\mathbf{S}^{(t)}_{N\times L}=P_\Omega(\mathbf{X}_{N\times L}-\mathbf{L}^{(t)}_{N\times L})$
where $P_\Omega(\cdot)$ is the projection of a matrix onto an entry set Ω.
End While

goals in the sense that this article primarily addresses prior target detection where the knowledge of targets to be detected is known as opposed to [51] which is devoted to anomaly detection where the targets to be detected are unknown. A second primary difference is that the detectors used for target detection are different. The detectors used in this article assume that the targets of interest are provided *a priori* such as CEM. By contrast, [51] assumes no knowledge at all about the targets such as the anomaly detector developed by Reed-Xiaoli in [22]. As a result, a third salient difference is that BA-OSP derived in this article annihilates BKG via OSP, while [51] made use of inverting the sample correlation matrix $\mathbf{R}$ to suppress BKG instead. A fourth major difference lies on the fact that OSP was used in [51] to re-interpret GoDec as an OSP version of Go-Dec, OSP-GoDec, but was not used to perform anomaly detection. In this article, OSP is particularly used to design BA-OSP to annihilate BKG and has nothing to do with the development of OSP-Go-Dec. Finally, the concept of data sphering developed in this article to derive various versions of OSP-based detectors was not developed in [51]. The purpose of using data sphering in BA-OSP is to remove the data statistics of the first and second orders in a similar manner that Reed-Xiaoli's anomaly detector inverts $\mathbf{R}$ to suppress BKG. This can remedy the issue in the leakage resulting from BKG suppression via performing data sphering on the subspaces generated by the low-rank and the sparse matrices.

III. DSOSP

OSP was originally developed from signal-to-noise ratio (SNR) and is a target detector which assumes that both the target to be detected specified by its spectral signature $\mathbf{d}$ and the undesired targets to be eliminated, specified by $p-1$ target signatures, $\mathbf{u}_1,\mathbf{u}_2,\ldots,\mathbf{u}_{p-1}$ are to be known *a priori*.

Instead of working on the original data matrix $\mathbf{X}$, we can pre-process $\mathbf{X}$ by data sphering to obtain a sphered data matrix $\hat{\mathbf{X}}$ prior to OSP. For those who are interested in data sphering, we refer its details to [2, Section VI-3]. Then, the $\mathbf{d}$ and $\mathbf{U}$ in the original data matrix $\mathbf{X}$ will be generated from $\hat{\mathbf{X}}$ as $\hat{\mathbf{d}}$ and $\hat{\mathbf{U}}$ to replace $\mathbf{d}$ and $\mathbf{U}$in the original data space $\mathbf{X}$, respectively, to be implemented in (2) to yield a DSOSP. More specifically, let $\hat{\mathbf{r}}$be the sphered data sample of $\mathbf{r}$. Then (2) can be modified and extended to

$$\delta^{\mathrm{DSOSP}}(\hat{\mathbf{r}})=\hat{\mathbf{d}}^T P^{\perp}_{\hat{\mathbf{U}}}\hat{\mathbf{r}}. \tag{10}$$

IV. OSP Using LRaSMD

This section presents ideas of how to implement OSP using LRaSMD with the low-rank matrix $\mathbf{L}$ and sparse matrix $\mathbf{S}$ generated by GoDec.

A. BA-OSP by Low-Rank Matrix L

The DSOSP derived in Section III operates the sphered data $\hat{\mathbf{X}}$ with an attempt to suppress BKG to some extent by removing the first-order and second-order data statistics. However, $\hat{\mathbf{X}}$ may still contain high-order statistics residuals of BKG, which cannot be removed by sphering. To further address this issue, the low-rank matrix $\mathbf{L}$ generated by LRaSMD from the original data matrix $\mathbf{X}$ can be used to describe BKG. In this case, we can use (1) to annihilate BKG by replacing $\mathbf{U}$ in (1) with BKG to derive a BKG-annihilated (BA) projector given by

$$P^{\perp}_{\mathbf{B}}=\mathbf{I}-\mathbf{B}\mathbf{B}^{\#}=\mathbf{I}-\mathbf{B}(\mathbf{B}^T\mathbf{B})^{-1}\mathbf{B}^T \tag{11}$$

where $\mathbf{B}$ is the matrix specified by BKG. Implementing the OSP in (2) coupled with $P^{\perp}_{\mathbf{B}}$ in (11) gives rise to an OSP with BKG-annihilated by $\mathbf{L}$, denoted by $\mathbf{L}$-OSP $\delta^{\mathrm{OSP}}_{\mathbf{L}}(\mathbf{r})$

$$\delta^{\mathrm{OSP}}_{\mathbf{B}}(\mathbf{r})=\mathbf{d}^T P^{\perp}_{\mathbf{U}}\left(P^{\perp}_{\mathbf{B}}\mathbf{r}\right)=\mathbf{d}^T P^{\perp}_{\mathbf{U}}\mathbf{r}^{\perp}_{\mathbf{B}}. \tag{12}$$

For example, if $\mathbf{B}$ in (12) is specified by the low-rank matrix $\mathbf{L}$, then $\mathbf{r}^{\perp}_{\mathbf{L}}=P^{\perp}_{\mathbf{L}}\mathbf{r}$ uses $\mathbf{L}$ to annihilate BKG.

B. Target-Specified OSP by Sparsity Matrix S With BA by L

The $\mathbf{L}$-OSP in (12) does not take into account the sparse matrix $\mathbf{S}$ which is assumed to contain target signal sources. To take full advantage of LRaSMD, an OSP, called ($\mathbf{L}$, $\mathbf{S}$)-OSP, denoted by $\delta^{\mathrm{OSP}}_{(\mathbf{L},\mathbf{S})}(\mathbf{r})$, can be derived as

$$\delta^{\mathrm{OSP}}_{(\mathbf{L},\mathbf{S})}(\mathbf{r})=\mathbf{d}^T_{\mathbf{S}}P^{\perp}_{\mathbf{U}_{\mathbf{S}}}\left(P^{\perp}_{\mathbf{L}}\hat{\mathbf{r}}\right)=\mathbf{d}^T_{\mathbf{S}}P^{\perp}_{\mathbf{U}_{\mathbf{S}}}\mathbf{r}^{\perp}_{\mathbf{L}} \tag{13}$$

where the subscript of ($\mathbf{L}$, $\mathbf{S}$) indicates that OSP operates the data space specified by the sparse matrix $\mathbf{S}$, while BKG is annihilated by the low-rank matrix $\mathbf{L}$. Particularly, $\mathbf{d}_{\mathbf{S}}$ and $\mathbf{U}_{\mathbf{S}}$ indicate that $\mathbf{d}$ and $\mathbf{U}$ are extracted from the subspace generated by the sparse matrix $\mathbf{S}$.

C. Target-Specified OSP by Low-Rank and Sparse Matrices L + S With BA by L

The ($\mathbf{L}$, $\mathbf{S}$)-OSP in (13) assumes that target signal sources are present in the data space generated by the sparse matrix $\mathbf{S}$, while discarding the fact that target signal sources may be also embedded in BKG generated by the low-rank matrix

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$\mathbf{L}$. To address this issue, an OSP, called $(\mathbf{L}, \mathbf{L}+\mathbf{S})$-OSP and denoted by $\delta^{\mathrm{OSP}}_{(\mathbf{L},\mathbf{L}+\mathbf{S})}(\mathbf{r})$ can be further derived by

$$\delta^{\mathrm{OSP}}_{(\mathbf{L},\mathbf{L}+\mathbf{S})}(\mathbf{r}) = \mathbf{d}^T_{\mathbf{L}+\mathbf{S}} P^{\perp}_{\mathbf{U}_{\mathbf{L}+\mathbf{S}}}\left(P^{\perp}_{\mathbf{L}}\mathbf{r}\right) = \mathbf{d}^T_{\mathbf{L}+\mathbf{S}} P^{\perp}_{\mathbf{U}_{\mathbf{L}+\mathbf{S}}}\mathbf{r}^{\perp}_{\mathbf{L}} \tag{14}$$

where the subscript of $(\mathbf{L}, \mathbf{L}+\mathbf{S})$ indicates that OSP operates data space specified by both the low-rank matrix $\mathbf{L}$ and sparse matrix $\mathbf{S}$ while BKG is annihilated by only the low-rank matrix $\mathbf{L}$. The $\mathbf{d}_{\mathbf{L}+\mathbf{S}}$, and $\mathbf{U}_{\mathbf{L}+\mathbf{S}}$ are specifically extracted from the subspace generated by the joint low-rank matrix $\mathbf{L}$ and the sparse matrix, $\mathbf{L}+\mathbf{S}$.

V. Jointly Using Data Sphering and LRaSMD for OSP

This section further develops various versions of OSP by combining DSOSP in Section III and OSP using LRaSMD in Section IV. Two matrices can be used to specify the target signal subspace for OSP. One is the sparse matrix $\mathbf{S}$ which contains target signal sources extracted from BKG. The other is $\mathbf{L}+\mathbf{S}$ which contains target signal sources embedded in BKG. Due to the fact that the target signal sources are generally described by high-order statistics greater than two, these matrices should be sphered prior to OSP as sphered $\mathbf{S}$, denoted by $\hat{\mathbf{S}}$ and sphered $(\mathbf{L}+\mathbf{S})$, denoted by $(\mathbf{L}\hat{+}\mathbf{S})$.

A. DSOSP With BA by L

The simplest combination of (12) and (13) is to implement DSOSP in (10) in conjunction with $\mathbf{L}$-OSP in (13), referred to as $\mathbf{L}$-DSOSP, denoted by $\delta^{\mathrm{DSOSP}}_{\mathbf{L}}(\mathbf{r})$ as

$$\delta^{\mathrm{DSOSP}}_{\mathbf{L}}(\mathbf{r}) = \hat{\mathbf{d}}^T P^{\perp}_{\hat{\mathbf{U}}}\left(P^{\perp}_{\mathbf{L}}\hat{\mathbf{r}}\right) = \hat{\mathbf{d}}^T P^{\perp}_{\hat{\mathbf{U}}}\hat{\mathbf{r}}^{\perp}_{\mathbf{L}} \tag{15}$$

where $\hat{\mathbf{d}}$ and $\hat{\mathbf{U}}$ are extracted from the sphered data matrix $\hat{\mathbf{X}}$, $\hat{\mathbf{r}}^{\perp}_{\mathbf{L}} = P^{\perp}_{\mathbf{L}}\hat{\mathbf{r}}$.

B. DSOSP With BA by $\hat{\mathrm{L}}$

In $\mathbf{L}$-DSOSP, the BKG is annihilated by the low-rank matrix $\mathbf{L}$ obtained directly from the original data matrix $\mathbf{X}$ via GoDec. This section slightly modifies $\mathbf{L}$-DSOSP by replacing $\mathbf{L}$ with $\hat{\mathbf{L}}$ which is the sphered $\mathbf{L}$. So, replacing $\mathbf{L}$ in (11) with $\hat{\mathbf{L}}$ yields $\hat{\mathbf{L}}$-DSOSP, denoted by $\delta^{\mathrm{DSOSP}}_{\hat{\mathbf{L}}}(\mathbf{r})$ and given as

$$\delta^{\mathrm{DSOSP}}_{\hat{\mathbf{L}}}(\mathbf{r}) = \hat{\mathbf{d}}^T P^{\perp}_{\hat{\mathbf{U}}}\left(P^{\perp}_{\hat{\mathbf{L}}}\hat{\mathbf{r}}\right) = \hat{\mathbf{d}}^T P^{\perp}_{\hat{\mathbf{U}}}\hat{\mathbf{r}}^{\perp}_{\hat{\mathbf{L}}} \tag{16}$$

where $\hat{\mathbf{d}}$ and $\hat{\mathbf{U}}$ are extracted from the sphered data matrix $\hat{\mathbf{X}}$ and $\hat{\mathbf{L}}$ is the sphered low-rank matrix $\mathbf{L}$.

C. Target-Specified OSP by Sparse Matrix S With BA by $\hat{\mathrm{L}}$

The detector presented in the following has its target signal subspace specified by the sparse matrix $\mathbf{S}$ and the BKG to be removed specified by the sphered low-matrix $\hat{\mathbf{L}}$. It is referred to as $(\hat{\mathbf{L}}, \mathbf{S})$-OSP, denoted by $\delta^{\mathrm{OSP}}_{(\hat{\mathbf{L}},\mathbf{S})}(\mathbf{r})$ as

$$\delta^{\mathrm{OSP}}_{(\hat{\mathbf{L}},\mathbf{S})}(\mathbf{r}) = \mathbf{d}^T_{\mathbf{S}} P^{\perp}_{\mathbf{U}_{\mathbf{S}}}\left(P^{\perp}_{\hat{\mathbf{L}}}\mathbf{r}\right) = \mathbf{d}^T_{\mathbf{S}} P^{\perp}_{\mathbf{U}_{\mathbf{S}}}\mathbf{r}^{\perp}_{\hat{\mathbf{L}}}. \tag{17}$$

D. Target-Specified OSP by $\hat{\mathrm{S}}$ With BA by L

In a complete opposite to (17) we can also specify the target signal subspace and BKG by $\hat{\mathbf{S}}$ and $\mathbf{L}$, respectively. The resulting detector is referred to as $(\mathbf{L}, \hat{\mathbf{S}})$-OSP, denoted by $\delta^{\mathrm{OSP}}_{(\mathbf{L},\hat{\mathbf{S}})}(\mathbf{r})$ as

$$\delta^{\mathrm{OSP}}_{(\mathbf{L},\hat{\mathbf{S}})}(\mathbf{r}) = \mathbf{d}^T_{\hat{\mathbf{S}}} P^{\perp}_{\mathbf{U}_{\hat{\mathbf{S}}}}\left(P^{\perp}_{\mathbf{L}}\mathbf{r}\right) = \mathbf{d}^T_{\hat{\mathbf{S}}} P^{\perp}_{\mathbf{U}_{\hat{\mathbf{S}}}}\mathbf{r}^{\perp}_{\mathbf{L}} \tag{18}$$

where $\hat{\mathbf{S}}$ is the sphered sparse matrix $\mathbf{S}$ and $\mathbf{d}$ extracted from $\hat{\mathbf{S}}$.

E. Target-Specified OSP by $\hat{\mathrm{S}}$ With $BA = \hat{\mathrm{L}}$

Now, if we specify the target signal subspace and BKG by $\hat{\mathbf{S}}$ and $\hat{\mathbf{L}}$, respectively, then we obtain $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP, denoted by $\delta^{\mathrm{OSP}}_{(\hat{\mathbf{L}},\hat{\mathbf{S}})}(\mathbf{r})$ as

$$\delta^{\mathrm{OSP}}_{(\hat{\mathbf{L}},\hat{\mathbf{S}})}(\mathbf{r}) = \mathbf{d}^T_{\hat{\mathbf{S}}} P^{\perp}_{\mathbf{U}_{\hat{\mathbf{S}}}}\left(P^{\perp}_{\hat{\mathbf{L}}}\mathbf{r}\right) = \mathbf{d}^T_{\hat{\mathbf{S}}} P^{\perp}_{\mathbf{U}_{\hat{\mathbf{S}}}}\mathbf{r}^{\perp}_{\hat{\mathbf{L}}} \tag{19}$$

where $\hat{\mathbf{S}}$ and $\hat{\mathbf{L}}$ are the sphered sparse matrix $\mathbf{S}$ and the sphered low-rank matrix $\mathbf{L}$, respectively, with $\mathbf{d}$ and $\mathbf{U}$ extracted from $\hat{\mathbf{S}}$.

F. Sphered Low-Rank and Sparse Matrices-Specified OSP by $(\mathrm{L}\hat{+}\mathrm{S})$ with BA by L

Compared to the previous BA-OSP detectors in (17)–(19) where the target signal subspace is specified by either $\mathbf{S}$ or $\hat{\mathbf{S}}$, the following BA-OSP detector specifies the target signal subspace by $(\mathbf{L}\hat{+}\mathbf{S})$, while the BKG is specified by $\mathbf{L}$. The resulting detector is referred to as $(\mathbf{L}, \mathbf{L}\hat{+}\mathbf{S})$-OSP, denoted by $\delta^{(\mathbf{L}\hat{+}\mathbf{S})\text{-OSP}}_{\mathbf{L}}(\mathbf{r})$as

$$\delta^{\mathrm{OSP}}_{(\mathbf{L},\mathbf{L}\hat{+}\mathbf{S})}(\mathbf{r}) = \mathbf{d}^T_{(\mathbf{L}\hat{+}\mathbf{S})} P^{\perp}_{\mathbf{U}_{(\mathbf{L}\hat{+}\mathbf{S})}}\left(P^{\perp}_{\mathbf{L}}\mathbf{r}\right) = \mathbf{d}^T_{(\mathbf{L}\hat{+}\mathbf{S})} P^{\perp}_{\mathbf{U}_{(\mathbf{L}\hat{+}\mathbf{S})}}\mathbf{r}^{\perp}_{\mathbf{L}}. \tag{20}$$

G. Sphered Low-Rank and Sparse Matrices-Specified OSP by $(\mathrm{L}\hat{+}\mathrm{S})$ With BA by $\hat{\mathrm{L}}$

Finally, we can have a BA-OSP detector to operate target signal and BKG subspaces on sphered $(\mathbf{L}\hat{+}\mathbf{S})$and $\hat{\mathbf{L}}$, respectively. The resulting detector is referred to as $(\mathbf{L}\hat{+}\mathbf{S}, \hat{\mathbf{L}})$-OSP, denoted by $\delta^{(\mathbf{L}\hat{+}\mathbf{S})\text{-OSP}}_{\hat{\mathbf{L}}}(\mathbf{r})$ as

$$\delta^{\mathrm{OSP}}_{(\hat{\mathbf{L}},\mathbf{L}\hat{+}\mathbf{S})}(\mathbf{r}) = \mathbf{d}^T_{(\mathbf{L}\hat{+}\mathbf{S})} P^{\perp}_{\mathbf{U}_{(\mathbf{L}\hat{+}\mathbf{S})}}\left(P^{\perp}_{\hat{\mathbf{L}}}\mathbf{r}\right) = \mathbf{d}^T_{(\mathbf{L}\hat{+}\mathbf{S})} P^{\perp}_{\mathbf{U}_{(\mathbf{L}\hat{+}\mathbf{S})}}\mathbf{r}^{\perp}_{\hat{\mathbf{L}}} \tag{21}$$

where $(\mathbf{L}\hat{+}\mathbf{S})$ is the sphered matrix $(\mathbf{L}+\mathbf{S})$ with $\mathbf{d}$ and $\mathbf{U}$ extracted from $(\mathbf{L}\hat{+}\mathbf{S})$. where $\mathbf{d}$ and $\mathbf{U}$ are extracted from $(\mathbf{L}\hat{+}\mathbf{S})$.

To summarize all the detectors developed in this article, Table I lists the eleven BA-OSP versions derived in Sections III–V, all of which will be used for experiments.

Fig. 1(a)–(k) also depicts the block diagrams of all BA-OSP versions in Table I with one-to-one correspondence between Fig. 1 and Table I.

VI. CEM

As noted in the introduction, the main goal of this article is to extend OSP to work as if it was CEM which inverts the correlation matrix $\mathbf{R}$ to suppress BKG. By taking advantage of the GoDec-generated LRaSMD, OSP can annihilate the BKG using (1). Therefore, to evaluate OSP along with its variants in Table I, CEM is a perfect candidate to be used for comparison.

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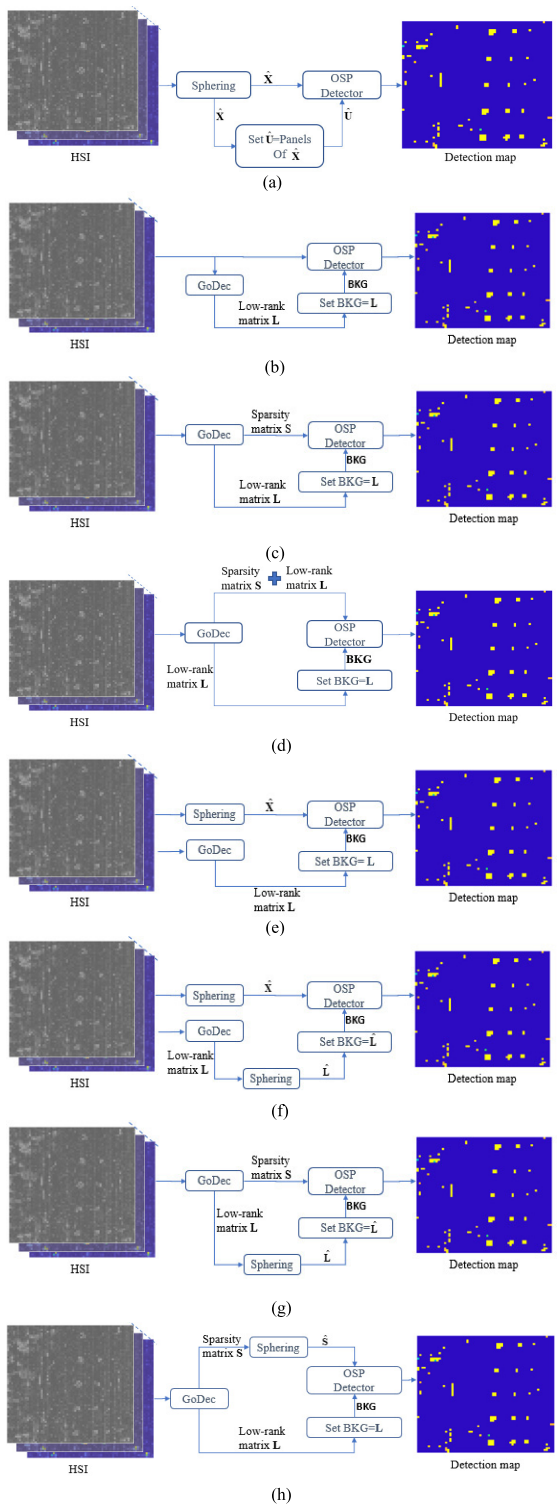


Fig. 1. Block diagrams of all BA-OSP versions in Table I. (a) Graphic diagram of implementing DSOSP. (b) Graphic diagram of implementing **L**-OSP. (c) Graphic diagram of (**L**, **S**)-OSP. (d) A graphic block diagram of (**L**, **L** + **S**)-OSP. (e) Block diagram of **L**-DSOSP. (f) Graphic diagram of $\hat{\mathbf{L}}$-DSOSP. (g) Graphic diagram of ($\hat{\mathbf{L}}$, **S**)-OSP. (h) Graphic diagram of (**L**, $\hat{\mathbf{S}}$)-OSP. (i) Graphic diagram of s($\hat{\mathbf{L}}$, $\hat{\mathbf{S}}$)-OSP. (j) Graphic diagram of (**L**, **L**$\hat{+}$**S**)-OSP. (k) Graphic diagram of (**L**$\hat{+}$**S**, $\hat{\mathbf{L}}$)-OSP.

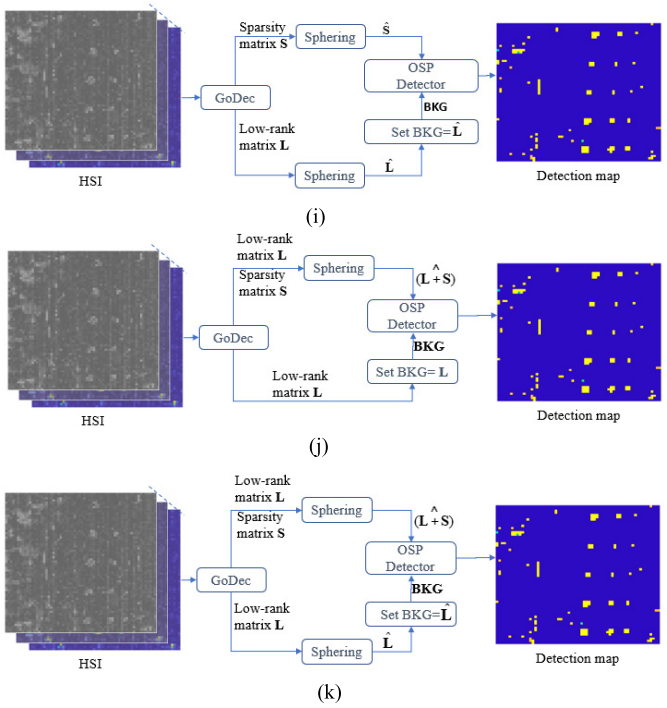


Fig. 1. *(Continued)* Block diagrams of all BA-OSP versions in Table I. (a) Graphic diagram of implementing DSOSP. (b) Graphic diagram of implementing **L**-OSP. (c) Graphic diagram of (**L**, **S**)-OSP. (d) A graphic block diagram of (**L**, **L** + **S**)-OSP. (e) Block diagram of **L**-DSOSP. (f) Graphic diagram of $\hat{\mathbf{L}}$-DSOSP. (g) Graphic diagram of ($\hat{\mathbf{L}}$, **S**)-OSP. (h) Graphic diagram of (**L**, $\hat{\mathbf{S}}$)-OSP. (i) Graphic diagram of s($\hat{\mathbf{L}}$, $\hat{\mathbf{S}}$)-OSP. (j) Graphic diagram of (**L**, **L**$\hat{+}$**S**)-OSP. (k) Graphic diagram of (**L**$\hat{+}$**S**, $\hat{\mathbf{L}}$)-OSP.

TABLE I
LIST OF BA-OSP VERSIONS USED FOR EXPERIMENTS

	BA-OSP versions	notation	Equation
(a)	DSOSP	$\delta^{\mathrm{DSOSP}}(\mathbf{r})$	(10)
(b)	**L**-OSP	$\delta^{\mathrm{OSP}}_{\mathbf{L}}(\mathbf{r})$	(12)
(c)	(**L**, **S**)-OSP	$\delta^{\mathrm{OSP}}_{(\mathbf{L},\mathbf{S})}(\mathbf{r})$	(13)
(d)	(**L**, **L**+**S**)-OSP	$\delta^{\mathrm{OSP}}_{(\mathbf{L},\mathbf{L}+\mathbf{S})}(\mathbf{r})$	(14)
(e)	**L**-DSOSP	$\delta^{\mathrm{DSOSP}}_{\mathbf{L}}(\mathbf{r})$	(15)
(f)	$\hat{\mathbf{L}}$ -DSOSP	$\delta^{\mathrm{DSOSP}}_{\hat{\mathbf{L}}}(\mathbf{r})$	(16)
(g)	($\hat{\mathbf{L}}$, **S**) -OSP	$\delta^{\mathrm{OSP}}_{(\hat{\mathbf{L}},\mathbf{S})}(\mathbf{r})$	(17)
(h)	(**L**, $\hat{\mathbf{S}}$) -OSP	$\delta^{\mathrm{OSP}}_{(\mathbf{L},\hat{\mathbf{S}})}(\mathbf{r})$	(18)
(i)	($\hat{\mathbf{L}}$, $\hat{\mathbf{S}}$) -OSP	$\delta^{\mathrm{OSP}}_{(\hat{\mathbf{L}},\hat{\mathbf{S}})}(\mathbf{r})$	(19)
(j)	(**L**, **L** $\hat{+}$ **S**) -OSP	$\delta^{\mathrm{OSP}}_{(\mathbf{L},\mathbf{L}\hat{+}\mathbf{S})}(\mathbf{r})$	(20)
(k)	($\hat{\mathbf{L}}$, **L** $\hat{+}$ **S**) -OSP	$\delta^{\mathrm{OSP}}_{(\hat{\mathbf{L}},\mathbf{L}\hat{+}\mathbf{S})}(\mathbf{r})$	(21)

Suppose that a hyperspectral image is represented by a collection of image pixel vectors, denoted by $\{\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N\}$ where $\mathbf{r}_i = (r_{i1}, r_{i2}, \ldots, r_{iL})^T$ for $1 \leq i \leq N$ is an L-dimensional pixel vector, N is the total number of pixels in the image and L is the total number of spectral channels. Further, assume that $\mathbf{d} = (d_1, d_2, \ldots, d_L)^T$ is specified by a desired signature of interest to be used for target detection. CEM is to find an FIR

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filter specified by a weight coefficient vector $\mathbf{w}$ that solves the following linearly constrained optimization problem:

$$\min_{\mathbf{w}}\{\mathbf{w}^T\mathbf{R}\mathbf{w}\} \text{ s.t. } \mathbf{d}^T\mathbf{w} = \mathbf{w}^T\mathbf{d} = 1. \tag{22}$$

where $\mathbf{R}_{L\times L} = (1/N)\sum_{i=1}^{N}\mathbf{r}_i\mathbf{r}_i^T$ is the auto-correlation sample matrix of $\{\mathbf{r}_i\}_{i=1}^{N}$. The optimal solution to (22), $\mathbf{w}^{\mathrm{CEM}}$ can be shown to be

$$\mathbf{w}^{\mathrm{CEM}} = \frac{\mathbf{R}_{L\times L}^{-1}\mathbf{d}}{\mathbf{d}^T\mathbf{R}_{L\times L}^{-1}\mathbf{d}}. \tag{23}$$

and CEM, denoted by $\delta^{\mathrm{CEM}}(\mathbf{r})$ is given by

$$\delta^{\mathrm{CEM}}(\mathbf{r}) = (\mathbf{w}^{\mathrm{CEM}})^T\mathbf{r} = (\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d})^{-1}(\mathbf{R}^{-1}\mathbf{d})^T\mathbf{r}. \tag{24}$$

By virtue of (24) CEM takes advantage of the sample correlation matrix $\mathbf{R}$ to perform BKG suppression via inverting $\mathbf{R}$ before it extracts the desired signature $\mathbf{d}$ where BKG is interpreted all signal sources other than the desired signature $\mathbf{d}$. With BKG suppression by $\mathbf{R}^{-1}$ followed by a matched filter using the signature $\mathbf{d}$ as its matched signal, CEM not only enhances its signal detectability, but also performs very effectively in suppressing impact resulting from unknown signal sources including interferers. Interestingly, using the inversion of $\mathbf{R}$ has the same effect as OSP using (1).

VII. 3-D ROC Curve-Derived Target Detection Measures

A commonly used measure for target detection is the ROCs analysis which is based on a curve plotted for detection probability (P_D) versus false alarm probability (P_F) [21] and the area under such a curve (AUC), referred to as A_Z, is used as a quantitative indicator to measure target detectability. Unfortunately, it is often the case that two detectors may have different ROC curves with the same A_Z. In other words, A_Z does not necessarily reflect the true target detectability of a detector. To address this issue, [2], [44]–[47] develops a 3-D ROC analysis which considers a 3-D ROC curve as a function of three parameters, P_D, P_F, and a threshold τ, and varying values of τ allows a user to generate a 3-D ROC curve. Most recently, a more detailed study on 3-D ROC analysis has been reported in [2], [46], [47]. In particular, it has been shown in [47] that a 2-D ROC curve of (P_D,P_F) can be only used to evaluate the effectiveness of a detector but does not necessarily represent P_D and P_F since both P_D and P_F are determined by the same threshold τ. As a matter of fact, in many cases, a 2-D ROC curve of (P_D,P_F) produced by a detector can be very high but its P_D indeed can be very low while its P_F can be also very high as well. A 3D ROC curve resolves these issues by expanding the traditional 2-D ROC (2-D ROC) curve of (P_D,P_F) to a 3-D ROC curve of (P_D,P_F,τ), from which three 2-D ROC curves, 2-D ROC curve of (P_D,P_F), 2-D ROC curve of (P_D,τ), and 2-D ROC curve of (P_F,τ) can be generated. By virtue of AUC values calculated from these three 2-D ROC curves, several new quantitative target detection measures derived in [47] can be used for our purpose.

A. Target Detection Measures

As noted in [47], $\mathrm{AUC}_{(\mathrm{D,F})}$ can be only used to evaluate the effectiveness of a detector, not its target detectability. To fully explore target detectability, we need to factor $\mathrm{AUC}_{(\mathrm{D},\tau)}$ into $\mathrm{AUC}_{(\mathrm{D,F})}$ to take care of both 2-D ROC curve of (P_D,P_F) and 2-D ROC curve of (P_D,τ) to derive a new target detectability, denoted by $\mathrm{AUC}_{\mathrm{TD}}$ as follows:

$$0 \le \mathrm{AUC}_{\mathrm{TD}} = \mathrm{AUC}_{(\mathrm{D,F})} + \mathrm{AUC}_{(\mathrm{D},\tau)} \le 2. \tag{25}$$

B. Joint Target Detection BKG Suppression Measure

P_D and P_F are two major criteria to evaluate the detection performance of a detector. According to the Neyman Pearson detection theory, it maximizes P_D by fixing P_F at a prescribed level. However, in hyperspectral imaging, BKG has a significant impact on target detection and plays a key role in detecting targets that may be embedded in BKG at subpixel scale. As a result, BKG suppression is crucial in hyperspectral target detection and can be specified by P_F. In this case, using a joint target detection and BKG suppression (TDBS) measure is more appropriate than using P_D and P_F alone and separately. To take both P_D and P_F into account, two detection measures can be derived as follows.

One is called BKG suppression measure defined by

$$-1 \le \mathrm{AUC}_{\mathrm{BS}} = \mathrm{AUC}_{(\mathrm{D,F})} - \mathrm{AUC}_{(\mathrm{F},\tau)}) \le 1. \tag{26}$$

The other is TDBS defined by

$$-1 \le \mathrm{AUC}_{\mathrm{TDBS}} = \mathrm{AUC}_{(\mathrm{D},\tau)} - \mathrm{AUC}_{(\mathrm{F},\tau)}) \le 1. \tag{27}$$

C. Signal-to-Noise Probability Ratio

As an alternative to $\mathrm{AUC}_{\mathrm{TDBS}}$ in (27), the concept of SNR which is widely used in communication can be also used to define a detection measure, called signal-to-noise probability ratio (SNPR), denoted by $\mathrm{AUC}_{\mathrm{SNPR}}$, as

$$0 \le \mathrm{AUC}_{\mathrm{SNPR}} = \frac{\mathrm{AUC}_{(\mathrm{D},\tau)}}{\mathrm{AUC}_{(\mathrm{F},\tau)}} \tag{28}$$

where the false alarm probability P_F can be assumed to be caused by noise.

D. Overall Detection Measure

In detection measures of TD in (25), BS in (26), TDBS in (27), and SNPR in (28), only two 2-D ROC curves are considered. This section takes into account AUC values of its corresponding three 2-D ROC curves [2-D ROC curves of (P_D,P_F), 2-D ROC curves of (P_D,τ), 2-D ROC curves of (P_F,τ)] to derive a new detection measure, called overall detection probability (ODP), $\mathrm{AUC}_{\mathrm{ODP}}$ defined by

$$-1 \le \mathrm{AUC}_{\mathrm{ODP}} = \mathrm{AUC}_{(\mathrm{D,F})} + \mathrm{AUC}_{(\mathrm{D},\tau)} - \mathrm{AUC}_{(\mathrm{F},\tau)} \le 2 \tag{29}$$

where $\mathrm{AUC}_{(\mathrm{F},\tau)}$ is subtracted from the sum of $\mathrm{AUC}_{(\mathrm{D,F})}$, $\mathrm{AUC}_{(\mathrm{D},\tau)}$. This is because higher values of $\mathrm{AUC}_{(\mathrm{D,F})}$ and $\mathrm{AUC}_{(\mathrm{D},\tau)}$ indicate higher detection performance. Conversely, a lower value of $\mathrm{AUC}_{(\mathrm{F},\tau)}$ indicates a better BKG suppression and thus, a better detection performance. As a result, (29)

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indeed combines the values calculated by the AUC values produced by the three 2-D ROC curves to produce a single quantitative value of the overall detection performance.

Interestingly, since $\mathrm{AUC}_{(\mathrm{F},\tau)}$ represents false alarm probability, $(1\text{-}\mathrm{AUC}_{(\mathrm{F},\tau)})$ will represent no-signal predictive probability in [28] or no-signal detectability. So, if we replace $\mathrm{AUC}_{(\mathrm{F},\tau)}$ by $(1\text{-}\mathrm{AUC}_{(\mathrm{F},\tau)})$ with $p(H_1)$ and $p(H_0)$ being denoted by prior probabilities of hypotheses, H_1 and H_0 in a binary hypothesis testing problem, we can derive a new detection measure

$$\mathrm{AUC}_{\mathrm{ODP}} = \mathrm{AUC}_{(\mathrm{D,F})} + p(H_1)\mathrm{AUC}_{(\mathrm{D},\tau)} + p(H_0)(1\text{-}\mathrm{AUC}_{(\mathrm{F},\tau)}) \tag{30}$$

with $0 \leq \mathrm{AUC}_{\mathrm{ODP}} \leq 3$ which can be also used to evaluate a combined performance of a detector for its effectiveness, target detectability, and no-signal detectability.

If we further assume that signal sources and BKG sources are considered as two different classes, denoted by signal class, C_{S} and BKG class, C_{BKG} along with their class sample ratios determined by $p_{\mathrm{S}} = (n_{\mathrm{S}}/N)$ and $p_{\mathrm{BKG}} = (n_{\mathrm{BKG}}/N)$, respectively, where n_{S}, n_{BKG}, and N are the number of data samples in C_{S} and C_{BKG} and the total number of data samples. Then (30) can be modified by letting $\mathrm{AUC}_{(\mathrm{D,F})}$ be zero and reduced to overall accuracy (OA) for a two-class (signal, BKG) classification problem defined by

$$0 \leq \mathrm{OA} = p_{\mathrm{S}}\mathrm{AUC}_{(\mathrm{D},\tau)} + p_{\mathrm{BKG}}(1\text{-}\mathrm{AUC}_{(\mathrm{F},\tau)}) \leq 1. \tag{31}$$

As a result, (31) can be considered as a special case of (30).

VIII. Image Data to be Studied for Experiments

A. HYDICE Data Scene

The image scene to be studied shown in Fig. 2 was acquired by the airborne hyperspectral digital imagery collection experiment (HYDICE) sensor in August 1995 from a flight altitude of 10000 ft. This scene has been studied extensively by many reports such as [1]–[4].

There are 15 square panels with three different sizes, 3 m × 3 m, 2 m × 2 m and 1 m × 1 m, respectively, shown in Fig. 2(a). Due to the approximately 1.56 m ground sampling distance, each of the panels in the first column except the first row contains two-panel pixels highlighted by red, p_{211}, p_{221} in row 2, p_{311}, p_{312} in row 3, p_{411}, p_{412} in row 4, p_{511}, p_{521} in row 5 as shown in Fig. 2(b) by ground truth. All the 11 remaining panels in Fig. 2(b) contain one single panel pixel for each panel also highlighted by red, p_{11}, p_{12}, p_{13} in row 1, p_{22}, p_{23} in row 2, p_{32}, p_{33} in row 3, p_{42}, p_{43} in row 4, p_{52}, p_{53} in row 5. Therefore, there are a total of 19-panel pixels. Fig. 2(b) shows their precise spatial locations with the pixels in yellow (Y pixels) indicating panel pixels mixed with the BKG. Fig. 2(c) plots the five-panel spectral signatures, $\mathbf{p}_1$, $\mathbf{p}_2$, $\mathbf{p}_3$, $\mathbf{p}_4$, $\mathbf{p}_5$ obtained from Fig. 2(b), where the ith panel signature, denoted by $\mathbf{p}_i$ was generated by averaging the red panel center pixels in row i.

According to [51], this same scene was also studied for OSP-GoDec. Instead of directly determining the values of m and k, it assumed that the ranks of the low-rank matrix $\mathbf{L}$ and sparse matrix $\mathbf{S}$ were specified by m and j, respectively. Then it used VD to estimate a parameter p which was the sum of r and j as $p = m + j$. This was followed by an algorithm, called MXSVD derived from the maximum orthogonal complement algorithm (MOCA) developed in [55] to estimate the value of j. Once the j is determined, both m and k can be obtained by $m = p - j$ and $k = j \times N$, respectively, where N is the total number of data samples. For those who are interested in detail, we refer to [56], [57].

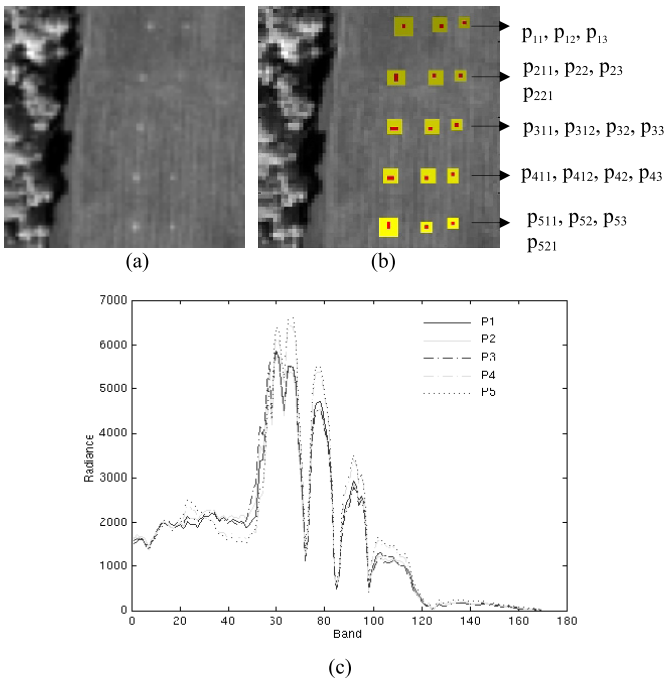


Fig. 2. (a) HYDICE panel scene which contains 15 panels. (b) Ground truth map of spatial locations of 19 R panel pixels. (c) Spectra of $\mathbf{p}_1$, $\mathbf{p}_2$, $\mathbf{p}_3$, $\mathbf{p}_4$, and $\mathbf{p}_5$.

B. AVIRIS San Diego Airport Scene

A second data set is an AVIRIS airport scene from San Diego, CA, USA as shown in Fig. 3(a). It has a size of 400 × 400 pixels with a 3.5 m spatial resolution and 224 spectral bands ranging from 370 to 2510 nm. Removing the bands corresponding to the water absorption regions, low SNR, and bad bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224) results in 189 bands. The area with a size of 100 × 100 pixels at the upper left corner of the scene marked by a red square was selected as the test image shown in Fig. 3(b) along with its ground-truth map shown in Fig. 3(c). It is an urban scene whose BKG is mainly made up of a roof, shadow, and grass. The targets of interest in the scene are three airplanes, which consist of 58 pixels and account for 0.33% of the test image as shown in Fig. 3(b). The spectral signatures of these airplanes were obtained from the ground truth in Fig. 3(c) and used as the desired signature $\mathbf{d}$.

Using the results in [51], Table II tabulates the value of p and m estimated by HFC with $\mathrm{P_F} \leq 10^{-3}$ and MX-SVD, $j = p - m$, and $k = j \times N$, and these values were used to execute GoDec. In addition, since GoDec uses q to overcome the singular value slow decay problem, q was set to, i.e., $q = 2$ [34] and $\varepsilon = 10^{-3}$ in the following experiments conducted in Section IX.

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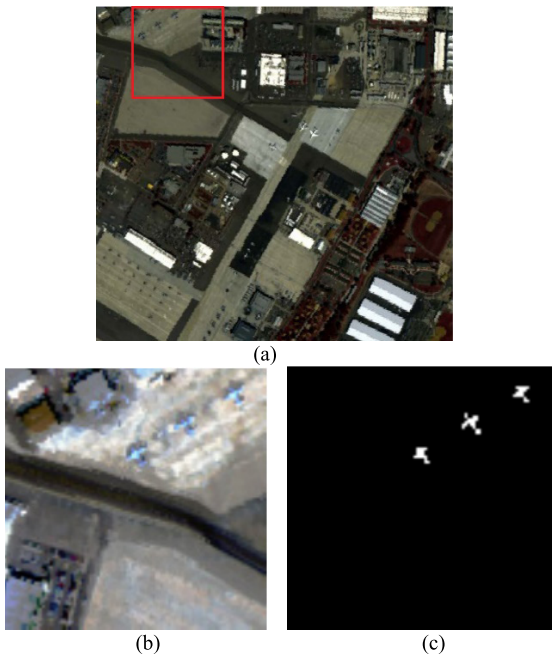


Fig. 3. AVIRIS San Diego Airport scene (a) pseudo-color image of the whole scene (b) pseudo-color image of the selected area (c) ground truth map.

TABLE II
ESTIMATION OF p AND m FOR HYDICE AND SAN DIEGO AIRPORT DATA SETS BY NWHFC USING MX-SVD

Data	VD	p	m by MX-SVD
HYDCE 15-panel scene	NWHFC $P_F \leq 10^{-3}$	13	7
San Diego airport scene	HFC, $P_F \leq 10^{-3}$	12	7

IX. EXPERIMENTS AND DISCUSSIONS

This section performs experiments by OSP with its various BA-OSP forms tabulated in Table I and CEM in Section V where one of five-panel signatures in Fig. 2(c) is designated as the desired target signature, $\mathbf{d}$ and the remaining four-panel signatures will be used as undesired target signatures, $\mathbf{U}$. The LRaSMD used in the experiments was produced by GoDec described in Section II. As noted at the end of Section II, OSP-GoDec in [51] can be also used for the same purpose. Since both produced very similar and close results, GoDec was used because of its popularity and its codes available on https://sites.google.com/site/godecomposition/code. So, those who are interested in our work can use it to verify our experimental results.

A. HYDICE Data Experiments

Figs. 4 and 5(a)–(k) show the detection maps produced by the original OSP in (2) and BA-OSP versions in Table I. As shown in Fig. 4 the original OSP performed poorly in detecting 19-panel pixels in Fig. 2 because the BKG is complicated and the $\mathbf{U}$ did not include BKG signatures. Fig. 5(a) shows the detection maps of 19-panel pixels in five rows by DSOSP which were amazingly improved by simply performing data sphering to clean up BKG prior to the implementation of OSP. This indicated that BKG could be indeed characterized by the first-order and second-order statistics. Fig. 5(b)–(d) further shows that using the low-rank matrix $\mathbf{L}$ as a BKG signature matrix for BKG annihilation was not good enough. Fig. 5(e) shows that the results obtained by DSOSP coupled with $\mathbf{L}$ were comparable to that by DSOSP in Fig. 5(a). Interestingly, if DSOSP implemented in conjunction with the sphered low-rank matrix, $\hat{\mathbf{L}}$, the detection results in Fig. 5(f) slightly improved the detection results in Fig. 5(a) and (e), specifically, the detection maps of the panel pixels in the second row. From Fig. 4, it is apparent that the best results were obtained by $(\mathbf{L}, \mathbf{L}\hat{+}\mathbf{S})$-OSP in Fig. 5(j) and $(\hat{\mathbf{L}}, \mathbf{L}\hat{+}\mathbf{S})$-OSP in Fig. 5(k) where BKG was annihilated by either $\mathbf{L}$ or $\hat{\mathbf{L}}$, while the target signal subspace was specified by the sphered joint matrix of $\mathbf{L}+\mathbf{S}$, $\mathbf{L}\hat{+}\mathbf{S}$. As shown in Fig. 5(j) and (k), the 19 R panel pixels were clearly detected, particularly, five subpixel panels in the third column. Most interestingly, the spatial locations of p_{211} and p_{221} vertically lined-up in row 2, p_{311} and p_{312} horizontally lined-up in row 3, p_{411} and p_{412} horizontally lined-up in row 4, p_{511} and p_{521} vertically lined-up in row five shown in Fig. 2(b) by ground truth were correctly detected but were not visualized in Fig. 5(a)–(i). This is because the panels are small targets and embedded in BKG. In this case, the joint matrix of $\mathbf{L}+\mathbf{S}$ must be used to represent the target signal subspace. However, since BKG is complex and may overwhelm the small panel pixels, data sphering was a necessary step to suppress BKG before the detection of panel pixels took place. For comparison, Fig. 6(a) and (b) also shows the detection maps of CEM and TCIMF which were comparable to the results of DSOSP in Fig. 5(a), $\mathbf{L}$-DSOSP in Fig. 5(e), and $\hat{\mathbf{L}}$-DSOSP in Fig. 5(f) with no visible differences. This made sense because CEM and TCIMF suppressed BKG by $\mathbf{R}^{-1}$ in the same way that OSP did a similar task by sphering the data. However, it also shows that using data sphering and low-rank matrix $\mathbf{L}$ to suppress BKG was obviously not sufficiently enough. This implies that the target signal subspace also played a key role in target detection, specifically, when there are subpixel and mixed pixels in BKG such as R panel pixels mixed with Y panel pixel in the first column in Fig. 2(b). To address this issue, both $\mathbf{L}$ and $\mathbf{S}$ matrices should be jointly considered as a target signal subspace. But even in this case, target signal sources may be still embedded in BKG. To uncover such target signal sources, data sphering was once again implemented to remove the first and second-order statistics from the space generated by $\mathbf{L}+\mathbf{S}$ to obtain its sphered data space, $\mathbf{L}\hat{+}\mathbf{S}$. The results are shown in Fig. 5(j)–(k) which were much better than that in Fig. 5(d) obtained by $(\mathbf{L}, \mathbf{L}+\mathbf{S})$-OSP without data sphering.

Figs. 4–6 only show qualitative results by visual inspection. To further conduct quantitative analysis, 3-D ROC curves along with their corresponding three 2-D ROC curves were generated from Figs. 4–6 to calculate $\text{AUC}_{(\text{D,F})}$, $\text{AUC}_{(\text{D},\tau)}$, $\text{AUC}_{(\text{F},\tau)}$ from which the detection measures derived in Section VII can be also calculated. Table III tabulates values of $\text{AUC}_{(\text{D,F})}$, $\text{AUC}_{(\text{D},\tau)}$, $\text{AUC}_{(\text{F},\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{TDBS}, AUC_{SNPR}, AUC_{ODP} where the best results are boldfaced. As we can see from Table III, $(\mathbf{L}, \hat{\mathbf{S}})$-OSP, $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP, $(\mathbf{L}, \mathbf{L}\hat{+}\mathbf{S})$-OSP, $(\hat{\mathbf{L}}, \mathbf{L}\hat{+}\mathbf{S})$-OSP were among the best target

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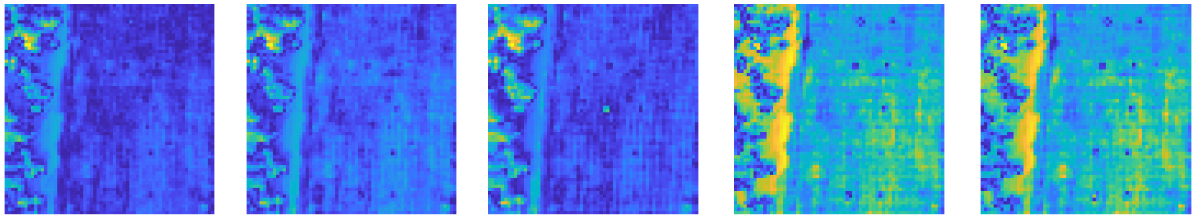


Fig. 4. Detection maps of OSP.

detectors, whereas OSP produced the worst results. In particular, $(\mathbf{L}, \hat{\mathbf{S}})$-OSP and $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP performed nearly the same and so did $(\mathbf{L}, \mathbf{L}\hat{+}\mathbf{S})$-OSP and $(\hat{\mathbf{L}}, \mathbf{L}\hat{+}\mathbf{S})$-OSP. Interestingly, these two sets of target detectors performed completely opposite in the sense of $\text{AUC}_{(\text{D,F})}$, $\text{AUC}_{(\text{D},\tau)}$, $\text{AUC}_{(\text{F},\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{TDBS}, AUC_{SNPR}, AUC_{ODP} in the detection of 19-panel pixels in five rows.

B. San Diego Airport Data Experiments

Same experiments conducted for the HYDICE 15-panel data scene were also performed for the San Diego airport scene in Fig. 3. Since the three airplanes are the only targets of interest to be detected in the scene and no prior knowledge is available to be used for $\mathbf{U}$, OSP, DSOSP, and TCIMF are not applicable to this scene. To address this particular issue in the unavailability of $\mathbf{U}$, BA-OSP is specifically developed for this purpose. It takes advantage of the GoDec-generated $\mathbf{L}$ and $\mathbf{S}$ to replace $\mathbf{U}$ to perform OSP. Fig. 7 shows the detection maps of (a) $\mathbf{L}$-OSP, (b) $(\mathbf{L}, \mathbf{S})$-OSP, (c) $(\mathbf{L}, \mathbf{L}+\mathbf{S})$-OSP, (d) $\mathbf{L}$-DSOSP, (e) $\hat{\mathbf{L}}$-DSOSP, (f) $(\hat{\mathbf{L}}, \mathbf{S})$-OSP, (g) $(\mathbf{L}, \hat{\mathbf{S}})$-OSP, (h) $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP, (i) $(\mathbf{L}, \mathbf{L}\hat{+}\mathbf{S})$-OSP, (j) $(\hat{\mathbf{L}}, \mathbf{L}\hat{+}\mathbf{S})$-OSP where it seemed that (g) $(\mathbf{L}, \hat{\mathbf{S}})$-OSP, (h) $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP, (i) $(\mathbf{L}, \mathbf{L}\hat{+}\mathbf{S})$-OSP, (j) $(\hat{\mathbf{L}}, \mathbf{L}\hat{+}\mathbf{S})$-OSP performed better than other BA-OSP detectors. However, such visual inspection is only for qualitative analysis. So, in analog with Table III, quantitative analysis is also further conduced. Table IV tabulates values of $\text{AUC}_{(\text{D,F})}$, $\text{AUC}_{(\text{D},\tau)}$, $\text{AUC}_{(\text{F},\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{TDBS}, AUC_{SNPR}, AUC_{ODP} calculated from the detection maps of Fig. 7 where the best results are boldfaced and were produced by (g) $(\mathbf{L}, \hat{\mathbf{S}})$-OSP, (h) $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP.

C. Discussions on BA-OSP

The experimental results presented in Figs. 4–7 and Tables III and IV provide many interesting observations and findings.

1) First of all, Tables III and IV validated a very important fact that solely relying on a single 2-D ROC curve of (P_D,P_F) to evaluate target detection performance as reported in hyperspectral target detection literature is not reliable. As shown in Table III $\text{AUC}_{(\text{D,F})}$ values were nearly 0.99 for CEM and all BA-OSP versions excluding the original OSP. Specifically, all the four target detectors, $(\mathbf{L}, \hat{\mathbf{S}})$-OSP, $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP, $(\mathbf{L}, \mathbf{L}\hat{+}\mathbf{S})$-OSP, $(\hat{\mathbf{L}}, \mathbf{L}\hat{+}\mathbf{S})$-OSP, produced $\text{AUC}_{(\text{D,F})} = 1$, in which case, there is no way to tell which one is better than another. Nevertheless, since $(\mathbf{L}, \hat{\mathbf{S}})$-OSP and $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP performed nearly the same and so did $(\mathbf{L}, \mathbf{L}\hat{+}\mathbf{S})$-OSP and $(\hat{\mathbf{L}}, \mathbf{L}\hat{+}\mathbf{S})$-OSP, it is suggested that either $(\mathbf{L}, \hat{\mathbf{S}})$-OSP or $(\mathbf{L}, \mathbf{L}\hat{+}\mathbf{S})$-OSP can be used as one of the best BA-OSP detectors.
2) Also, according to Table III, (g) $(\mathbf{L}, \hat{\mathbf{S}})$-OSP, (h) $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP, (i) $(\mathbf{L}, \mathbf{L}\hat{+}\mathbf{S})$-OSP, (j) $(\hat{\mathbf{L}}, \mathbf{L}\hat{+}\mathbf{S})$-OSP were the best among all the versions of BA-OSP listed in Table I. According to visual inspection of the results in Fig. 5, $(\mathbf{L}, \mathbf{L}\hat{+}\mathbf{S})$-OSP, $(\hat{\mathbf{L}}, \mathbf{L}\hat{+}\mathbf{S})$-OSP, and CEM seemed to perform slightly better than $(\mathbf{L}, \hat{\mathbf{S}})$-OSP and $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP did. Specifically, in the detection of panels in row 4 the former had better BKG suppression than the latter as also verified by $\text{AUC}_{(\text{F},\tau)}$ in Table III. However, this conclusion was reversed for target detectability shown by $\text{AUC}_{(\text{D},\tau)}$ in Table III. Furthermore, if both BKG suppression and target detectability are jointly taken into account in terms of TDBS, SNPR, ODP, the overall performance in detection of 15 panels in five rows of $(\mathbf{L}, \hat{\mathbf{S}})$-OSP and $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP seemed to be little better than that of $(\mathbf{L}, \mathbf{L}\hat{+}\mathbf{S})$-OSP and $(\hat{\mathbf{L}}, \mathbf{L}\hat{+}\mathbf{S})$-OSP within negligible differences. Nevertheless, these results suggested that BKG suppression is more crucial than target detectability in terms of visual inspection. Such BKG suppression could clean up BKG interference on target detection and was accomplished by data spahering in the BA-OSP in (g–j). Similar observations can be also made on Fig. 7 and Table IV. Interestingly, in this case, (g) $(\mathbf{L}, \hat{\mathbf{S}})$-OSP, (h) $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP performed slightly better than (i) $(\mathbf{L}, \mathbf{L}\hat{+}\mathbf{S})$-OSP, (j) $(\hat{\mathbf{L}}, \mathbf{L}\hat{+}\mathbf{S})$-OSP.
3) As shown by experiments, a simple data sphering can significantly improve the OSP performance and performs as well as CEM and TCIMF can in Fig. 6. Including data sphering in OSP makes OSP work as if it was CEM inverting the correlation matrix $\mathbf{R}$ to suppress BKG. In other words, that OSP performs data sphering is similar to that CEM performs $\mathbf{R}^{-1}$.
4) Introducing LRaSMD into OSP significantly enhances target detection capability where the low-rank matrix $\mathbf{L}$ and sparse matrix $\mathbf{S}$ play crucial roles in target detection of OSP. Also, including $\mathbf{L}$ into OSP allows OSP to annihilate BKG so as to derive BA-OSP. In addition, constraining target signal sources in the sphered target subspace generated by $\mathbf{S}$ or $\mathbf{L} + \mathbf{S}$ can further enhance target detectability.

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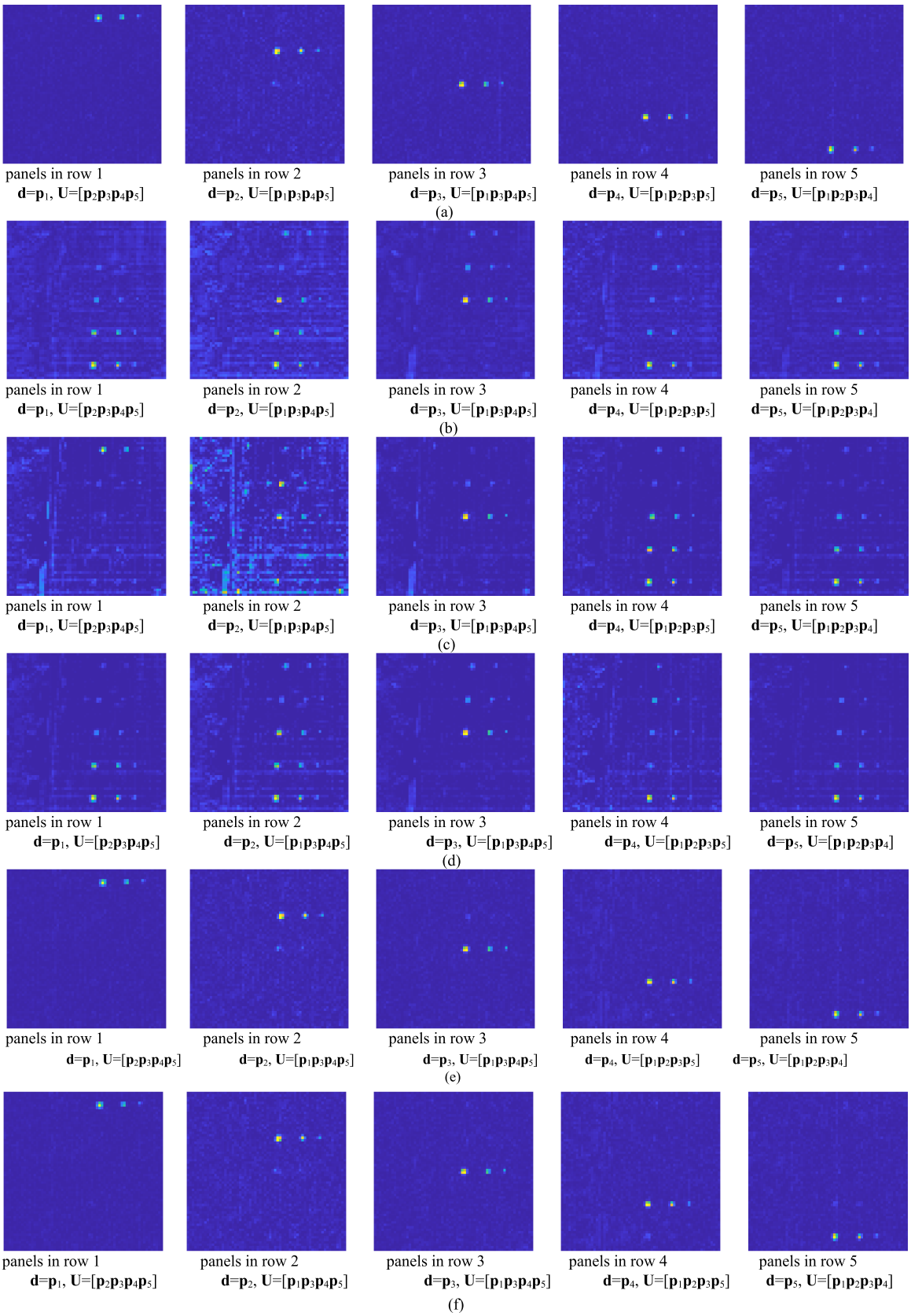


Fig. 5. Detection maps of panels in five rows by OSP and various versions of BA-OSP in Table I. Detection maps of (a) DSOSP, (b)$\mathbf{L}$-OSP, (c) $(\mathbf{L}, \mathbf{S})$-OSP, (d) $(\mathbf{L}, \mathbf{L}+\mathbf{S})$-OSP, (e) $\mathbf{L}$-DSOSP, (f) $\hat{\mathbf{L}}$-DSOSP, (g) $(\hat{\mathbf{L}}, \mathbf{S})$-OSP, (h) $(\mathbf{L}, \hat{\mathbf{S}})$-OSP and BKG $= \mathbf{L}$, (i) $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP, (j) $(\mathbf{L}, \mathbf{L} \dotplus \mathbf{S})$-OSP, and (k) $(\hat{\mathbf{L}}, \mathbf{L} \dotplus \mathbf{S})$-OSP.

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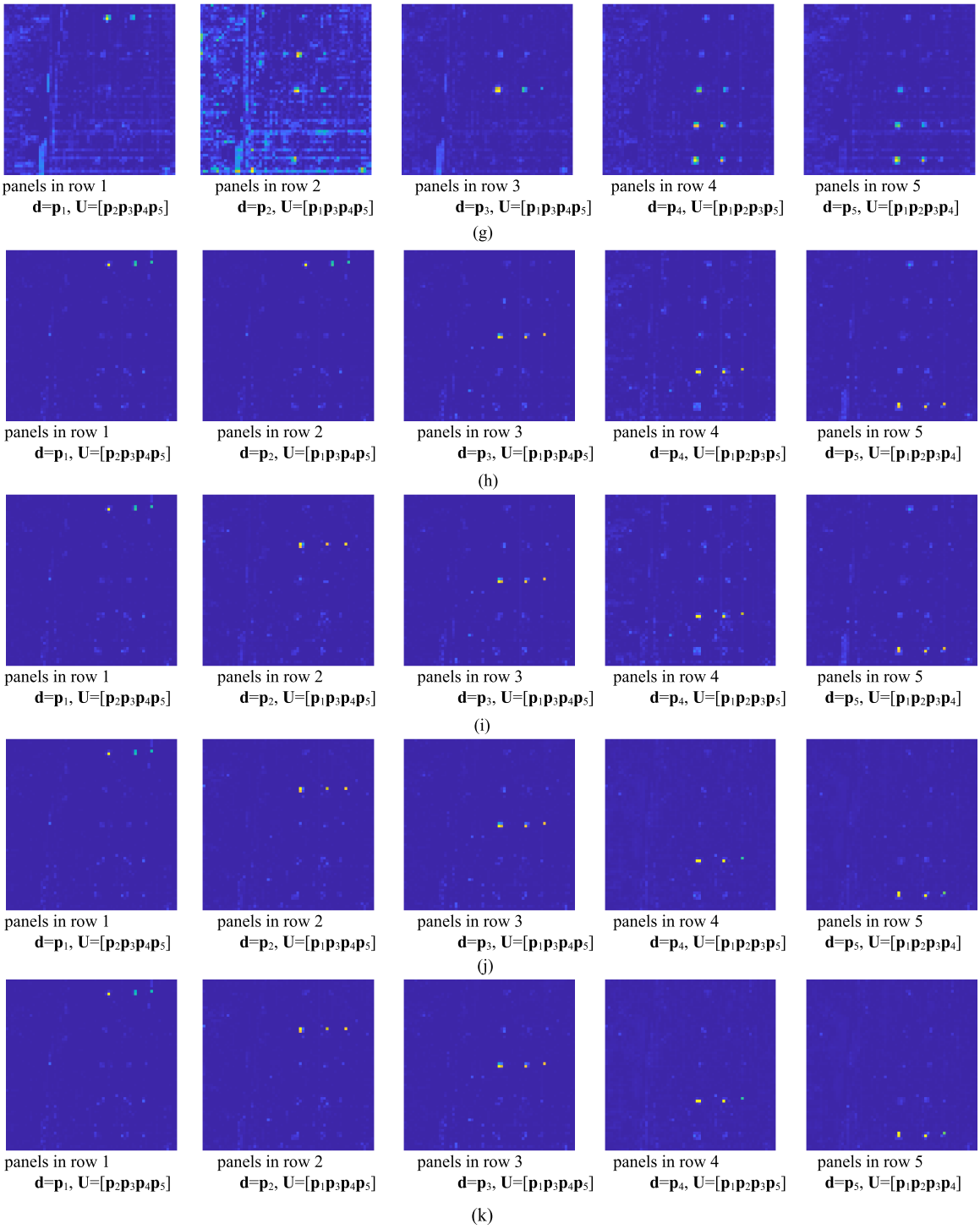


Fig. 5. *(Continued)* Detection maps of panels in five rows by OSP and various versions of BA-OSP in Table I. Detection maps of (a) DSOSP, (b) **L**-OSP, (c) (**L**, **S**)-OSP, (d) (**L**, **L** + **S**)-OSP, (e) **L**-DSOSP, (f) $\hat{\mathbf{L}}$-DSOSP, (g) ($\hat{\mathbf{L}}$, **S**)-OSP, (h) (**L**, $\hat{\mathbf{S}}$)-OSP and BKG = **L**, (i) ($\hat{\mathbf{L}}$, $\hat{\mathbf{S}}$)-OSP, (j) (**L**, **L**$\hat{+}$**S**)-OSP, and (k) ($\hat{\mathbf{L}}$, **L**$\hat{+}$**S**)-OSP.

5) Joint implementation of data sphering and LRaSMD generally performs better than implementing one of them alone.
6) It should be noted that the HYDICE panel data set in Fig. 2 provides the complete ground truth of targets of interest with different sizes to be detected in the complex BKG, i.e., 3 m × 3 m visible two-pixel pure panels in the first column, 2 m × 2 m nearly visible one pixel pure panels in the second column mixed with BKG and 1 m × 1 m invisible subpixel panels in the third column embedded in BKG. This 15-panel HYDICE data set presents different challenging levels in target detection.

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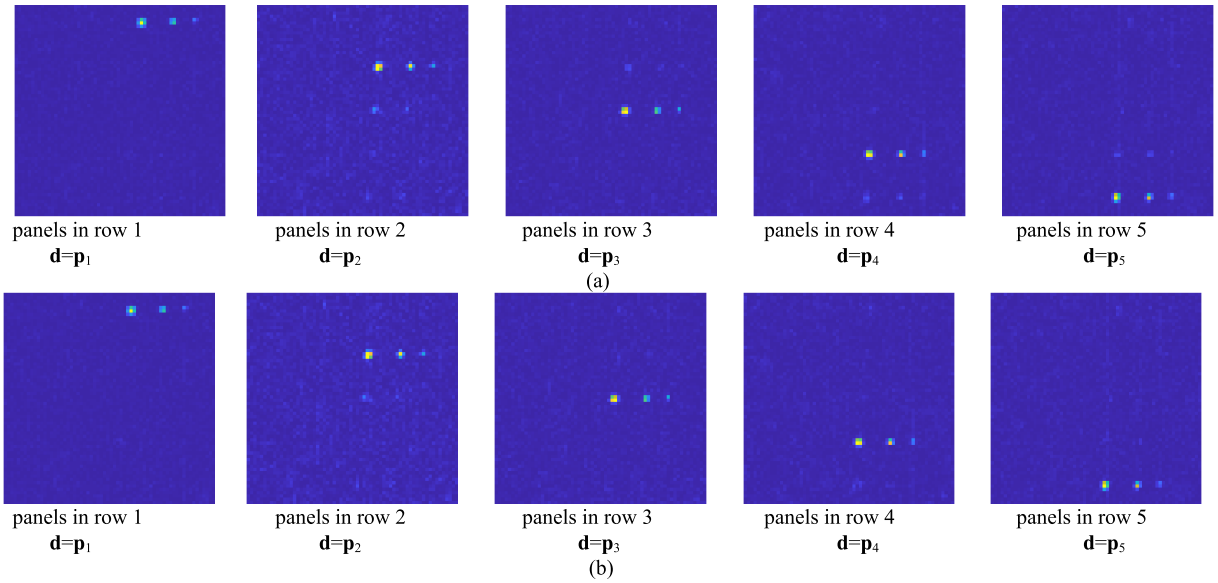


Fig. 6. Detection maps of panels in five rows by CEM and TCIMF (a) CEM and (b) TCIMF.

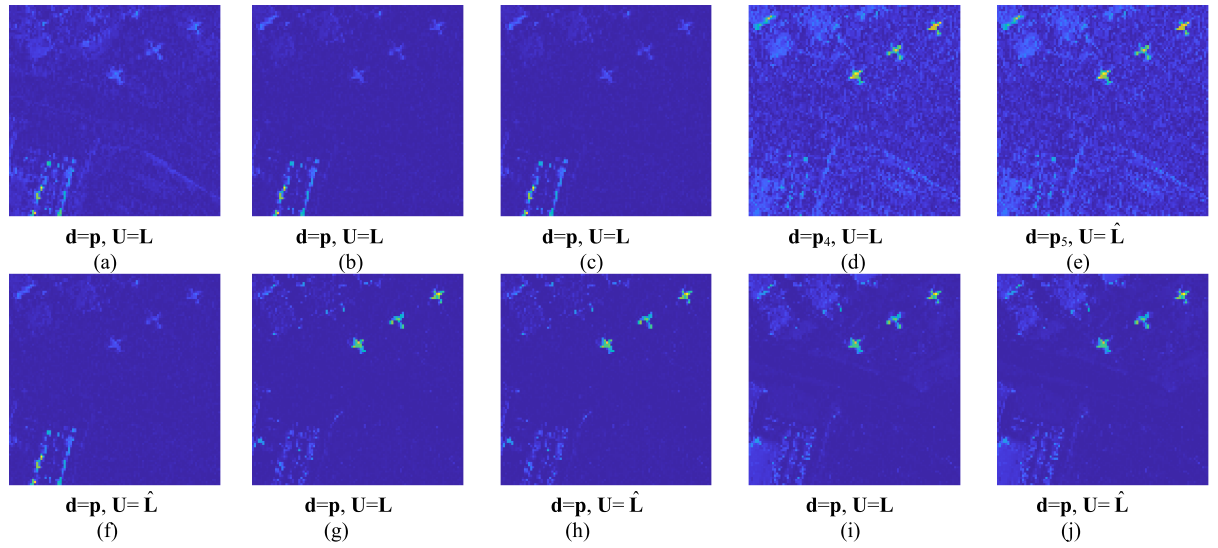


Fig. 7. Detection maps of (a) **L**-OSP, (b) (**L**, **S**)-OSP, (c) (**L**, **L** + **S**)-OSP, (d) **L**-DSOSP, (e) $\hat{\mathbf{L}}$-DSOSP, (f) ($\hat{\mathbf{L}}$, **S**)-OSP, (g) (**L**, $\hat{\mathbf{S}}$)-OSP, (h) ($\hat{\mathbf{L}}$, $\hat{\mathbf{S}}$)-OSP, (i) (**L**, **L**$\hat{+}$**S**)-OSP, and (j) ($\hat{\mathbf{L}}$, **L**$\hat{+}$**S**)-OSP.

It allows us to conduct detailed studies and analyses of how different scenarios of BA-OSP worked. Despite that many other data sets can be used for experiments they may not have sufficient target characteristics like HYDICE panel pixels to justify the effectiveness of target detection. For example, the targets of interest in the San Diego airport scene in Fig. 3 are the three airplanes which are too big to miss and easy to be detected. Besides, unlike the HYDICE data which has subpixel and mixed targets, this scene does not have such small targets of interest with ground truth that can be used to verify their presence. Accordingly, the San Diego airport scene is less interesting than the HYDICE data.

7) Finally, it should be noted that all the conducted experiments used the GoDec-generated LRaSMD model. Nevertheless, the same experiments can be also carried out for LRaSMD generated from OSP-GoDec [51]. The same conclusions can be also drawn. In this case, their results are not included.

X. Complexity Analysis on Comparison with Other Recent Developed Target Detection Techniques

There are many target detection methods reported in the literature. Depending upon targets to be detected target detection can be also categorized in [3] into prior target detection which requires prior knowledge of targets of interest and anomaly detection which requires no target knowledge at all. In other words, prior target detection and anomaly detection are completely separate concepts. For example, the prior

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TABLE III
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROWS DETECTION RESULTS

d	detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}
$\mathbf{p}_1$	CEM	0.9993	0.5953	0.0084	1.5946	0.9905	0.5869	70.8589	1.5862
	TCIMF	0.9993	0.5985	0.0084	1.5951	0.9905	0.5874	70.9681	1.5867
	OSP	0.4220	0.1056	0.1431	0.5276	0.2788	-0.0376	0.7375	0.3844
	DSOSP	0.9993	0.5953	0.0085	1.5945	0.9904	0.5868	70.2480	1.5861
	L-OSP	0.9362	0.0772	0.0311	1.0134	0.9049	0.0461	2.4841	0.9824
	(**L**,**S**)-OSP	0.9973	0.5728	0.0220	1.5702	0.9739	0.5508	26.0063	1.5481
	(**L**, **L**+**S**)-OSP	0.9780	0.0780	0.0125	1.0560	0.9653	0.0655	6.2444	1.0435
	L-DSOSP	0.9993	0.5957	0.0090	1.5949	0.9899	0.5867	66.4924	1.5860
	$\hat{\mathbf{L}}$ -DSOSP	0.9993	0.5957	0.0090	1.5949	0.9899	0.5867	66.4924	1.5860
	$(\hat{\mathbf{L}}, \mathbf{S})$ -OSP	0.9973	0.5728	0.0220	1.5702	0.9739	0.5508	26.0063	1.5481
	$(\mathbf{L}, \hat{\mathbf{S}})$ -OSP	**1.0000**	**0.7781**	**0.0036**	**1.7781**	**0.9964**	**0.7745**	**215.1948**	**1.7745**
	$(\hat{\mathbf{L}}, \hat{\mathbf{S}})$ -OSP	**1.0000**	**0.7781**	**0.0036**	**1.7781**	**0.9964**	**0.7745**	**215.1948**	**1.7745**
	$(\mathbf{L}, \mathbf{L} \dot{+} \mathbf{S})$ -OSP	**1.0000**	0.7555	**0.0036**	1.7555	**0.9964**	0.7519	208.1115	1.7519
	$(\hat{\mathbf{L}}, \mathbf{L} \dot{+} \mathbf{S})$ -OSP	**1.0000**	0.7555	**0.0036**	1.7555	**0.9964**	0.7519	208.1115	1.7519
$\mathbf{p}_2$	CEM	0.9998	0.8325	0.0169	1.8323	0.9828	0.8156	49.2400	1.8154
	TCIMF	0.9998	0.8322	0.0178	1.8320	0.9819	0.8114	46.7095	1.8142
	OSP	0.1258	0.0963	0.2235	0.2221	-0.0977	-0.1272	0.4308	-0.0014
	DSOSP	0.9998	0.8306	0.0178	1.8305	0.9820	0.8129	46.6863	1.8127
	L-OSP	0.9720	0.1527	0.0469	1.1246	0.9249	0.1058	3.2574	1.0778
	(**L**,**S**)-OSP	0.9589	0.5824	0.0604	1.5413	0.8982	0.5220	9.6390	1.4809
	(**L**, **L**+**S**)-OSP	0.9890	0.1584	0.0203	1.1474	0.9684	0.1381	7.7946	1.1271
	L-DSOSP	0.9998	0.8276	0.0187	1.8274	0.9811	0.8089	44.3589	1.8087
	$\hat{\mathbf{L}}$ -DSOSP	0.9998	0.8276	0.0187	1.8274	0.9811	0.8089	44.3589	1.8087
	$(\hat{\mathbf{L}}, \mathbf{S})$ -OSP	0.9589	0.5824	0.0604	1.5413	0.8982	0.5220	9.6390	1.4809
	$(\mathbf{L}, \hat{\mathbf{S}})$ -OSP	**1.0000**	**0.9148**	0.0048	**1.9148**	0.9952	**0.9100**	190.6705	**1.9100**
	$(\hat{\mathbf{L}}, \hat{\mathbf{S}})$ -OSP	**1.0000**	**0.9148**	0.0048	**1.9148**	0.9952	**0.9100**	190.6705	**1.9100**
	$(\mathbf{L}, \mathbf{L} \dot{+} \mathbf{S})$ -OSP	**1.0000**	0.9063	**0.0046**	1.9063	**0.9954**	0.9017	**196.4913**	1.9017
	$(\hat{\mathbf{L}}, \mathbf{L} \dot{+} \mathbf{S})$ -OSP	**1.0000**	0.9063	**0.0046**	1.9063	**0.9954**	0.9017	**196.4913**	1.9017
$\mathbf{p}_3$	CEM	0.9997	0.7658	0.0108	1.7655	0.9888	0.7550	70.8975	1.7547
	TCIMF	0.9997	0.7599	0.0117	1.7596	0.9879	0.7482	64.9999	1.7479
	OSP	0.6651	0.4188	0.2163	1.0839	0.4487	0.2025	1.9364	0.8676
	DSOSP	0.9997	0.7581	0.0116	1.7578	0.9879	0.7464	65.0982	1.7461
	L-OSP	0.9996	0.7494	0.0183	1.7490	0.9812	0.7310	40.8636	1.7307
	(**L**,**S**)-OSP	0.9996	0.7403	0.0123	1.7399	0.9872	0.7280	60.1569	1.7276
	(**L**, **L**+**S**)-OSP	0.9996	0.7157	0.0096	1.7153	0.9899	0.7062	74.8312	1.7057
	L-DSOSP	0.9997	0.7571	0.0120	1.7568	0.9875	0.7451	62.8593	1.7448
	$\hat{\mathbf{L}}$ -DSOSP	0.9997	0.7571	0.0120	1.7568	0.9875	0.7451	62.8593	1.7448
	$(\hat{\mathbf{L}}, \mathbf{S})$ -OSP	0.9996	0.7403	0.0123	1.7399	0.9872	0.7280	60.1569	1.7276
	$(\mathbf{L}, \hat{\mathbf{S}})$ -OSP	**1.0000**	0.8999	**0.0041**	1.8999	**0.9959**	0.8958	**218.8781**	1.8958
	$(\hat{\mathbf{L}}, \hat{\mathbf{S}})$ -OSP	**1.0000**	0.8999	**0.0041**	1.8999	**0.9959**	0.8958	**218.8781**	1.8958
	$(\mathbf{L}, \mathbf{L} \dot{+} \mathbf{S})$ -OSP	**1.0000**	**0.9079**	0.0046	**1.9079**	0.9954	**0.9033**	199.0058	**1.9033**
	$(\hat{\mathbf{L}}, \mathbf{L} \dot{+} \mathbf{S})$ -OSP	**1.0000**	**0.9079**	0.0046	**1.9079**	0.9954	**0.9033**	199.0058	**1.9033**
$\mathbf{p}_4$	CEM	0.9998	0.8149	0.0104	1.8147	0.9894	0.8045	78.5316	1.8044
	TCIMF	0.9998	0.8136	0.0113	1.8134	0.9885	0.8032	72.1416	1.8021
	OSP	0.2248	0.2784	0.4260	0.5032	-0.2012	-0.1475	0.6537	0.0773
	DSOSP	0.9998	0.8136	0.0114	1.8134	0.9883	0.8022	71.2645	1.8020
	L-OSP	0.9642	0.2675	0.0283	1.2317	0.9356	0.2391	9.4434	1.2033
	(**L**,**S**)-OSP	0.9978	0.6681	0.0133	1.6659	0.9843	0.6548	50.3225	1.6526

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TABLE III

(Continued) AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROWS DETECTION RESULTS

	(**L**, **L**+**S**)-OSP	0.8649	0.1191	0.0191	0.9840	0.8455	0.1000	6.2283	0.9649
	L-DSOSP	0.9998	0.8109	0.0149	1.8107	0.9848	0.7960	54.3740	1.7958
	$\hat{\mathbf{L}}$ -DSOSP	0.9998	0.8109	0.0149	1.8107	0.9848	0.7960	54.3740	1.7958
	$(\hat{\mathbf{L}}, \mathbf{S})$ -OSP	0.9978	0.6681	0.0133	1.6659	0.9843	0.6548	50.3225	1.6526
	$(\mathbf{L}, \hat{\mathbf{S}})$ -OSP	**1.0000**	**0.9501**	0.0082	**1.9501**	0.9918	**0.9419**	115.9907	**1.9419**
	$(\hat{\mathbf{L}}, \hat{\mathbf{S}})$ -OSP	**1.0000**	**0.9501**	0.0082	**1.9501**	0.9918	**0.9419**	115.9907	**1.9419**
	$(\mathbf{L}, \mathbf{L} \dot{+} \mathbf{S})$ -OSP	**1.0000**	0.9347	**0.0072**	1.9347	**0.9928**	0.9275	**129.7096**	1.9275
	$(\hat{\mathbf{L}}, \mathbf{L} \dot{+} \mathbf{S})$ -OSP	**1.0000**	0.9347	**0.0072**	1.9347	**0.9928**	0.9275	**129.7096**	1.9275
$\mathbf{p}_5$	CEM	0.9998	0.7376	0.0103	1.7373	0.9894	0.7273	71.4553	1.7270
	TCIMF	0.9997	0.7355	0.0111	1.7353	0.9886	0.7244	66.2089	1.7242
	OSP	0.5189	0.4433	0.4232	0.9621	0.0956	0.0201	1.0475	0.5390
	DSOSP	0.9998	0.7353	0.0111	1.7350	0.9885	0.7241	65.9661	1.7239
	L-OSP	0.9987	0.7056	0.0204	1.7043	0.9778	0.6852	34.5081	1.6838
	(**L**,**S**)-OSP	0.9989	0.7020	0.0131	1.7009	0.9854	0.6889	53.6338	1.6878
	(**L**, **L**+**S**)-OSP	0.9987	0.7062	0.0099	1.7050	0.9884	0.6963	71.3612	1.6951
	L-DSOSP	0.9998	0.7345	0.0133	1.7343	0.9864	0.7212	55.2489	1.7210
	$\hat{\mathbf{L}}$ -DSOSP	0.9998	0.7345	0.0133	1.7343	0.9864	0.7212	55.2489	1.7210
	$(\hat{\mathbf{L}}, \mathbf{S})$ -OSP	0.9989	0.7020	0.0131	1.7009	0.9854	0.6889	53.6338	1.6878
	$(\mathbf{L}, \hat{\mathbf{S}})$ -OSP	**1.0000**	0.9134	**0.0061**	1.9134	**0.9939**	0.9073	**149.8469**	1.9073
	$(\hat{\mathbf{L}}, \hat{\mathbf{S}})$ -OSP	**1.0000**	0.9134	**0.0061**	1.9134	**0.9939**	0.9073	**149.8469**	1.9073
	$(\mathbf{L}, \mathbf{L} \dot{+} \mathbf{S})$ -OSP	**1.0000**	**0.9386**	0.0063	**1.9386**	0.9937	**0.9323**	149.3720	**1.9323**
	$(\hat{\mathbf{L}}, \mathbf{L} \dot{+} \mathbf{S})$ -OSP	**1.0000**	**0.9386**	0.0063	**1.9386**	0.9937	**0.9323**	149.3720	**1.9323**

NOTE: **L**: LOW RANK, **S**: SPARSE MATRIX, $\hat{\mathbf{L}}$: SPHERED LOW-RANK MATRIX, $\hat{\mathbf{S}}$: SPHERED SPARSE MATRIX, DSOSP: DATA SPHERED OSP

TABLE IV

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROWS DETECTION RESULTS OBTAINED FROM FIG. 7

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}
(a) **L**-OSP	0.9875	0.6578	0.0691	1.6453	0.9183	0.5887	9.5254	1.5763
(b) (**L**,**S**)-OSP	0.9841	0.6528	0.0400	1.6370	0.9439	0.6129	16.3387	1.5970
(c) (**L**, **L**+**S**)-OSP	0.9854	0.6528	0.0400	1.6382	0.9452	0.6129	16.3395	1.5983
(d) **L**-DSOSP	0.9863	0.6220	0.0479	1.6083	0.9381	0.5741	12.9739	1.5604
(e) $\hat{\mathbf{L}}$ -DSOSP	0.9863	0.6220	0.0479	1.6083	0.9381	0.5741	12.9739	1.5604
(f) $(\hat{\mathbf{L}}, \mathbf{S})$ -OSP	0.9841	0.6528	0.0400	1.6370	0.9439	0.6129	16.3387	1.5970
(g) $(\mathbf{L}, \hat{\mathbf{S}})$ -OSP	0.9880	**0.7092**	**0.0172**	**1.6971**	**0.9706**	**0.6920**	**41.2905**	**1.6800**
(h) $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$ -OSP	0.9880	**0.7092**	**0.0172**	**1.6971**	**0.9706**	**0.6920**	**41.2905**	**1.6800**
(i) $(\mathbf{L}, \mathbf{L} \dot{+} \mathbf{S})$ -OSP	0.9839	0.7060	0.0242	1.6899	0.9594	0.6818	29.1753	1.6657
(j) $(\hat{\mathbf{L}}, \mathbf{L} \dot{+} \mathbf{S})$ -OSP	0.9839	0.7060	0.0242	1.6899	0.9594	0.6818	29.1753	1.6657

NOTE: **L**: LOW RANK, **S**: SPARSE MATRIX, $\hat{\mathbf{L}}$: SPHERED LOW-RANK MATRIX, $\hat{\mathbf{S}}$: SPHERED SPARSE MATRIX, DSOSP: DATA SPHERED OSP

target detection is generally performed in a supervised or an unsupervised manner to obtain the desired target knowledge. By contrast, anomaly detection is carried out without appealing for any target knowledge. Accordingly, their design rationales are also quite different. In this article, we place the main focus on the prior target detection and leave anomaly detection to another topic to be investigated in the future.

As for the prior target detection it can be also classified into spectral detection which only utilizes spectral information to perform target detection and spatial/spectral detection which makes use of both spectral and spatial information to perform target detection.

In early developments for spectral target detection various spectral measures have been used to discriminate one target from another [2, Chapter 16], such as spectral angle mapper [1], spectral information divergence (SID) [58], adaptive coherence estimator (ACE) [59], [60] and then followed by spectral detectors [2, Chapter 2], such as

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Gaussian statistics-based detectors, matched subspace detector (MSD) [61], adaptive subspace detector (ASD) [62], and SNR-based OSP [5]. One major issue arising in the above-mentioned target detection is the BKG factor which generally has a significant impact on target detection if the prior target knowledge is insufficient or targets of interest embedded in BKG as subpixel targets. To address the BKG issue CEM in [29] was believed to be the first work to use a concept derived from adaptive beamforming in [63] to suppress BKG by inverting the sample correlation matrix $\mathbf{R}$. The key difference between OSP and CEM is that OSP must know BKG as $\mathbf{U}$ and annihilate BKG via $P_{\mathbf{U}}^{\perp}$ in (2), while CEM does not have to know BKG for its annihilation like what OSP does but rather suppress the entire image as BKG by $\mathbf{R}^{-1}$ in (24). Another approach was to propose a piecewise linear strategy (PLS) for target detection, which is based on the assumption that the desired target is generally linearly separable from each BKG object [64]. By taking advantage of both OSP and CEM, an approach called TCIMF was further derived in [65] to not only annihilate undesired target signatures as $\mathbf{U}$ by $P_{\mathbf{U}}^{\perp}$ but also suppress BKG by $\mathbf{R}^{-1}$. By taking a different route from CEM and TCIMF which suppress unknown BKG, the BA-OSP presented in this article annihilates BKG and operates data samples on their respective subspaces specified by LRaSMD, a task which accomplishes BKG annihilation and targets to be detected in the LRaSMD-specified subspaces simultaneously. Nevertheless, the work presented in this article indeed lays a foundation for future research directions such as extending BA-OSP to nonlinear kernel-based methods, such as kernel OSP [66], kernel ASD [67], kernel MSD [68], kernel OSP [68], kernel CEM [69], and kernel TCIMF [70]. In particular, an approach similar to BA-OSP is currently being investigated by using TCIMF as a multiple target detector coupled with LRaSMD for hyperspectral multiple-target detection.

Most recently, several approaches have been investigated by including spatial information into spectral detectors to better characterize BKG for its estimation or reconstruction to avoid target samples from being compromised or overwhelmed by BKG. However, such spatial/spectral target detection generally results in a network with complicated structures and suffers from three major issues. One is that it may not work effectively for subpixel targets which are generally invisible with the unavailability of spatial information and also has no spatial correlation with its surrounding data samples. Another is the use of training samples to train a network to extract features. A third and the most serious issue is model complexity which has a sizable parameter set needed to be tuned by trial and error. It requires significant time in doing so. If they are tuned unproperly, the results are far worse than those reported. For example, [71] developed a deep spatial-spectral network (DSSN) which detected a region of interest map by an edge-preserving filter through training samples and then used the hyperspectral image and the corresponding RoI map as its inputs to extract spatial and spectral feature of interest via training samples that were used by the nearest neighbors (NNs) for detection-map refinement. Unfortunately, it requires ten parameters to be tuned to implement DSSN, which are the spatial size of training patch N_p, σ_r, and σ_s of domain transform recursive filter, λ_1 and λ_2 of the loss function, learning rate, size of the extracted spatial and spectral features of interest in spectral dimension K, reduction ratio r, minibatch size N_b, and kernel size along with the spectral dimension k_D. Shi *et al.* [72] developed a 3-D macro-micro residual autoencoder (3-D MMRAE) model which describes spectral information, and spatial information in macro and micro scales to extract features that will be used for a follow-up hierarchical radial basis function (hRBF) detector for target preservation and BKG suppression. Upon implementing 3-D-MMRAE there are four main parameters, including the macro patch size, the micro patch size, the feature dimension K, and the parameter σ in the distance constraint. Zhang *et al.* [73] developed a binary hypothesis model based on sparse representation (SRBBH) where two hypotheses are represented by BKG only subspace and the target with BKG subspace through training samples so that detection is made by comparing the reconstruction residuals under the different hypotheses. To implement SRBBH, there are parameters that need to be tuned, dual window sizes, outer window region (OWR) and inner window region (IWR), selection of BKG training samples, sparsity levels for two hypotheses, K_0 and K_1.

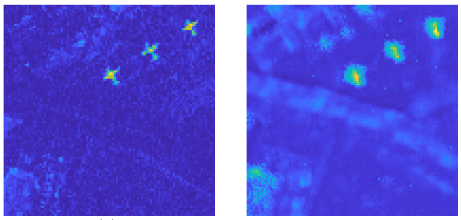


Fig. 8. Detection maps of (a) CEM and (b) DSSN.

So, comparing spatial/spectral target detection methods described above, BA-OSP has great advantages in several aspects, (1) simple and easy implementation; (2) no parameters needed to be tuned; (3) no training samples required; and (4) no need of BKG estimation or reconstruction, all of which are crucial in practical applications. Most importantly, BA-OSP enhances the strengths of OSP and CEM in detection performance via data sphering and LRaSMD whose parameters, p, m, j can be automatically determined by VD and MXSVD.

In order to demonstrate the advantages of BA-OSP we further conducted a comparative study and analysis with one of the most recently developed approaches, DSSN in [71], and the one of most popular detectors, CEM. Fig. 8 shows the detection maps of CEM and DSSN for qualitative analysis by visual inspection. By comparing the detection maps obtained by various versions of BA-OSP in Fig. 7 it clearly showed that (g) $(\mathbf{L}, \hat{\mathbf{S}})$-OSP, (h) $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP, (i) $(\mathbf{L}, \mathbf{L}\hat{+}\mathbf{S})$-OSP, (j) $(\hat{\mathbf{L}}, \mathbf{L}\hat{+}\mathbf{S})$-OSP in Fig. 7 performed comparably to CEM and outperformed DSSN in the sense of detecting three airplanes as well as suppressing BKG at the left bottom corner.

For a further quantitative study and analysis, Table V also tabulates values of $\mathrm{AUC}_{(\mathrm{D,F})}$, $\mathrm{AUC}_{(\mathrm{D},\tau)}$, $\mathrm{AUC}_{(\mathrm{F},\tau)}$, $\mathrm{AUC}_{\mathrm{TD}}$, $\mathrm{AUC}_{\mathrm{BS}}$, $\mathrm{AUC}_{\mathrm{TDBS}}$, $\mathrm{AUC}_{\mathrm{SNPR}}$, $\mathrm{AUC}_{\mathrm{ODP}}$. In comparison with the results in Table IV, the values of $\mathrm{AUC}_{(\mathrm{D,F})}$ of CEM and DSSN were very high as reported in [71] and also higher than

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TABLE V
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROWS DETECTION RESULTS OBTAINED FROM FIG. 8

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}
CEM	0.9955	0.6260	0.0395	1.6215	0.9559	0.5865	15.8288	1.5820
DSSN	0.9957	0.5848	0.1094	1.5805	0.8862	0.4754	5.3471	1.4711

those in Table IV. However, if we looked into other detection measures, $AUC_{(D,\tau)}$, $AUC_{(F,\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{TDBS}, AUC_{SNPR}, AUC_{ODP}, DSSN actually performed poorly on these measures, specifically compared to (g) $(\mathbf{L}, \hat{\mathbf{S}})$-OSP, (h) $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP in Table IV which were the best cases among all BA-OSP versions. The only detection measure that DSSN performed better than (g) $(\mathbf{L}, \hat{\mathbf{S}})$-OSP and (h) $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP is $AUC_{(D,F)}$. Interestingly, comparing the detection map of DSSN in Fig. 8(b) against Fig. 7(g)–(h), the three airplanes and BKG detected by DSSN were blurred. Why would its value of $AUC_{(D,F)}$ be higher than any other detectors? This is because $AUC_{(D,F)}$ is calculated by the 2-D ROC curve using the same threshold where a higher P_D also yields a higher P_F. This is why DSSN did very poor BKG suppression. This can be witnessed by the fact that DSSN produced $AUC_{(F,\tau)} = 0.1094$ almost ten times higher than 0.01072 produced by (g) $(\mathbf{L}, \hat{\mathbf{S}})$-OSP and (h) $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP. Also, even for the case of target detectability, DSSN produced $AUC_{(D,\tau)} = 0.5848$ which was lower than 0.7092 produced by (g) $(\mathbf{L}, \hat{\mathbf{S}})$-OSP and (h) $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP. Most impressively, (g) $(\mathbf{L}, \hat{\mathbf{S}})$-OSPand (h) $(\hat{\mathbf{L}}, \hat{\mathbf{S}})$-OSP. $AUC_{SNPR} = 41.2905$ which is almost 8 times higher than $AUC_{SNPR} = 5.3471$ produced by DSSN. Accordingly, it is misleading in evaluating target detection solely based on a single value of $AUC_{(D,F)}$.

XI. CONCLUSION

OSP has been found in many applications for hyperspectral data exploitation. However, its capability is also constrained by the necessity of prior knowledge of undesired signatures $\mathbf{U}$. It is known that BKG has a significant effect on hyperspectral target detection and has also tremendous impact on the OSP performance. Unfortunately, obtaining complete or reliable BKG knowledge to be used as $\mathbf{U}$ for OSP is nearly impossible. The main contribution of this article is to develop three approaches to addressing this issue. One is data sphering to remove the first- and second-order statistics so as to achieve BKG suppression prior to OSP. Another is BA-OSP which takes advantage of the GoDec-generated LRaSMD to find the low-rank matrix $\mathbf{L}$ so that it can be used as $\mathbf{U}$ to remove BKG. A third approach is to jointly combine data sphering and LRaSMD to perform OSP. Finally, the utilization of 3-D ROC curve-derived detection measures is fully explored in the conducted experiments to show that these three approaches can significantly improve OSP in target detection.

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